The Benefits of Delay to Online Decision-Making

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Real-time decisions are usually irrevocable in many contexts of online decision-making. One common practice is delaying real-time decisions so that the decision-maker can gather more information to make better decisions. For example, in online retailing, there is typically a time delay between when an online order is received and when it gets picked and assembled for shipping. However, decisions cannot be delayed forever. In this paper, we study this fundamental trade-off and aim to theoretically characterize the benefits of delaying real-time decisions. We provide a theoretical foundation for a broad family of online decision-making problems by proving that the gap between our proposed online algorithm (called "delayed Bayesian prophet") and the offline optimal hindsight policy decays exponentially fast in the length of delay. We also conduct extensive numerical experiments on the benefits of delay, using both synthetic data and publicly available real data. Both our theoretical and empirical results demonstrate an important managerial insight: a little delay is all we need. Finally, we extend our analysis and results to the setting where the arrival distribution is independent but non-identical, the setting where the arrival distribution is unknown, and the setting where decisions are made in batches.

Key words: online decision-making, resource allocation, delay, batching, multisecretary problem, order fulfillment, multi-item order

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1. Introduction

Real-time decisions are irrevocable in many contexts of online decision-making, from online retailing to ride-hailing. These decisions are challenging to make due to the uncertainty of demand and supply. In practice, these decisions are often delayed for a certain duration before being committed, which reaps many benefits. Indeed, when a large-scale e-tailer like Amazon or Alibaba receives an online order, there is typically a delay between when the order is received and when it gets dispatched for picking and packing. This delay

brings benefits such as maintaining a backlog to smooth out the workload and the ability to prioritize urgent orders (e.g., Xu et al. 2009). Meanwhile, in ride-hailing, Uber has implemented a delayed dispatching algorithm that matches riders with drivers in batches. As a result, overall wait times have gone down compared to Uber's previous matching algorithm, which seemed faster because it immediately matched riders with the closest available driver (Uber 2022). This wait-and-match idea has been applied in various other contexts, including kidney exchange (Ashlagi and Roth 2021) and online video games (Chen et al. 2021). As another non-conventional example, in tech companies like Amazon, newly hired junior research scientists are often paired with senior research scientists who act as mentors. Due to differences in research interests, group dynamics, and other factors, individuals may have varying preferences. Each senior research scientist can mentor only a limited number of junior scientists. In addition, the dynamic nature of tech companies often leads to scientists frequently joining or leaving the organization. Rather than matching a junior scientist with a mentor immediately upon their arrival, delaying the match by a few weeks or even months can be beneficial.

Unsurprisingly, delaying decisions also causes externalities on the system. In e-commerce, delays translate to slower delivery times and can cause warehouse congestion ahead of an outbound deadline. In matching markets and video game matchmaking, delays lead to long wait times for individual agents, who may then leave the platform. Therefore, it is paramount to understand the benefits gained from a specific duration of allowable delay, unlocking a rich set of questions depending on the online decision.

Overall, the benefits of delaying real-time decisions can be placed into the following categories.

1. **Work backlog**: Delaying the processing of jobs allows for maintaining a backlog of tasks, which can be necessary logistically (see Graves 2022), and also allows workers to choose the sequence of tasks (see Ibanez et al. 2018).
2. **Market thickness**: This is the benefit of increasing the range of feasible matches when both supply and demand in a market are arriving online. For example, in ride-hailing, delaying a rider's request allows for additional drivers to become available during that time, leading to a better match on average and reducing the overall wait time. The aforementioned settings where all participants arrive online (Uber 2022, Ashlagi and Roth 2021, Chen et al. 2021) fall under this category.

3. **Holding rewards**: This is the explicit benefit of holding individual orders in certain settings. For example, it gives customers a chance to cancel before dispatch in online retailing (Wu 2018) and ride-hailing, and allows for delaying orders for consolidation in e-commerce fulfillment (Wei et al. 2021, Aminian et al. 2023, Chen et al. 2024, Wang et al. 2024).
4. **Informational**: This is the benefit of gathering more advance information in order to make better online decisions. For example, in order fulfillment, decision-makers can obtain more accurate information about subsequent orders by delaying the decision on the current order, enabling them to make more informed warehouse assignments.

This paper focuses on the fourth and final of these benefits, which is perhaps also the most subtle and surprising, as the only gain is the ability of the online algorithm to make better decisions. Indeed, it has been observed in simulations that delaying order fulfillment decisions even slightly can reap significant benefits for an online retailer with a two-layer network (Zhao et al. 2020). It has also been observed in e-commerce practice that delays need not be long (Wang et al. 2024). Generally, these phenomena have not been well understood, and our paper fills this gap in the literature by providing a theoretical justification for these observations.

Specifically, we study an online allocation problem with a constraint on the length of delay, in which resources with fixed initial budgets are given, and the decision for each arriving demand is delayed by exactly the allowable length. Our model (**Section 3**) covers a wide range of practical applications across diverse fields such as e-commerce, manufacturing, and recruitment. For example, in manufacturing, production scheduling decisions can be optimized by incorporating allowable delays, enabling the integration of real-time production data and machine availability. Similarly, in the recruitment process, hiring decisions can be enhanced by permitting delays to accommodate further candidate assessments. More generally, for online allocation problems with additional delay costs, investigating the benefits of delay enables us to understand the balance between the benefits and costs of delay, helping decision-makers determine the optimal delay length.

In terms of results, we leverage recent literature starting with Vera and Banerjee (2021), Vera et al. (2021), Banerjee and Freund (2024) that establishes a constant upper bound on the regret of a "Bayesian prophet" algorithm for online resource allocation. We propose a natural extension of the Bayesian prophet algorithm called "delayed Bayesian prophet,"

where every demand is processed exactly K time steps later, and show that its regret bound decays exponentially in K. This extended framework leads to sharper constants than Vera and Banerjee (2021) (see Remark 1), and our presentation also offers corrections to Vera and Banerjee (2021) while being arguably simpler (see Lemma 1 and its proof). We study the multisecretary special case (**Section 4**) before considering a generalized online assignment problem (**Section 5**). We also extend to the non-IID (independent and identically distributed) setting (**Section 6.1**) and the IID unknown setting (**Section 6.2**), and analyze the performance of batching (**Section 6.3**). Finally, we conduct extensive numerical experiments on the benefits of delay, using both synthetic data (**Sections 7.1 and 7.2**) and publicly available real data (**Section 7.3**). Both our theoretical and empirical results demonstrate the important managerial insight that a little delay is all we need — further increasing the length of delay has diminishing marginal returns.

2. Literature Review

Online order fulfillment: Although we study a fairly large class of online resource allocation problems, our main motivation is the online fulfillment problem. Xu et al. (2009) were the first to study the online fulfillment optimization problem, and there has been a considerable amount of research on this topic in recent years, including Acimovic and Graves (2015), Jasin and Sinha (2015), Andrews et al. (2019), Zhao et al. (2020), DeValve et al. (2023), Amil et al. (2023), and Ma (2023). We refer to a tutorial on the fulfillment optimization problem by Acimovic and Farias (2019). A new research trend in e-commerce fulfillment is middle-mile delivery and we refer interested readers to recent papers by Chen et al. (2020) and Feng et al. (2024).

Job batching in order fulfillment and other contexts: Xu et al. (2009) examine the practical benefits of reactively batching orders by analyzing real data. Wei et al. (2021) study the optimal time to delay fulfillment and how to proactively batch orders. Order holding is also studied for different purposes: to consolidate multiple orders placed by the same person within a short amount of time or to catch order cancellations (e.g., Wu 2018, Aminian et al. 2023, Chen et al. 2024, Wang et al. 2024), whereas in this paper, orders are held to gather more information for making better fulfillment decisions.

We refer interested readers to Gurvich and Ward (2014), Ashlagi et al. (2019), Akbarpour et al. (2020), Ashlagi and Roth (2021), and Feng and Niazadeh (2024) for batching in

other contexts, such as two-sided online marketplaces that include ride-hailing and kidney exchange. There are at least three differences between batching in the context of order fulfillment and in the context of two-sided marketplaces. First, in order fulfillment, each decision is delayed by a certain amount of time, whereas the decisions are made in batches in online marketplaces. Second, there is a fundamental trade-off between increasing market thickness and mitigating the risk that participants may abandon the market. Jobs can become obsolete in two-sided marketplaces, whereas jobs never expire in order fulfillment. Third, supply is fixed and cannot be replenished in our context of resource allocation, whereas in two-sided marketplaces, both supply and demand have new arrivals.

Recently Kerimov et al. (2023, 2024) introduced the notion of all-time regret to allow for some delay in online matching while penalizing excessive delay. Wei et al. (2023) have since extended this concept to *multi-way* matching, where more than two jobs can be "matched" (consolidated) together for a larger reward. Wei et al. (2023) also consider an IID unknown setting where the arrival rates of different types are unknown a priori.

Finally, it is worth noting that delaying real-time decisions is equivalent to having advance demand information, which has been studied in the contexts of inventory replenishment (e.g., Gallego and Özer 2001, Lu et al. 2003) and queueing (e.g., Xu and Chan 2016). For the single-server admission control problem, there is a substantial understanding of the amount of future information necessary to achieve a convergent queue length (Spencer et al. 2014, Xu 2015).

We note that Golrezaei and Yao (2021) have recently studied a similar online resource allocation problem where some customers are patient and willing to wait in a buffer as long as they are guaranteed service. The authors derive competitive ratio guarantees for a (time-expanded) single-leg revenue management problem under adversarial arrivals, whereas we focus on regret guarantees for a multi-resource allocation problem under stochastic arrivals.

Constant regret in online resource allocation problems: Vera and Banerjee (2021) is the most relevant work to ours. The authors study a large class of online resource allocation problems and develop a general technique (called the Bayes selector) that can lead to an online policy with constant expected regret, i.e., the additive loss with respect to (w.r.t.) the offline benchmark is independent of the number of arrivals and resource levels. Vera and Banerjee (2021) generalize prior and contemporaneous results (e.g., Reiman and Wang 2008, Jasin and Kumar 2012, Arlotto and Gurvich 2019, Bray 2019, Bumpensanti

and Wang 2020, Vera et al. 2021, Banerjee and Freund 2024). In our paper, we generalize their model and technique to the setting with delay.

Unknown distribution in online resource allocation problems: Precise knowledge of the distribution (either demand or reward) is usually unavailable in online allocation, and the decision-maker is required to learn the distribution from realized observations while optimizing real-time decisions. We refer interested readers to Jasin (2015), Jiang et al. (2020), Li et al. (2020), Vera et al. (2021), Balseiro et al. (2023), and Li et al. (2024) for various unknown distributions/inputs in the context of online allocation. In our paper, we consider the simpler setting of unknown *stationary* demand, for which constant regret can be established, and we also investigate how this constant shrinks with delay.

3. Model

In this section, we describe our model. Let us consider the following finite-horizon discrete-time online resource allocation problem. There are M different types of resources indexed by $i \in [M] \triangleq \{1, \ldots, M\}$, and the initial budget vector is $B = (B_1, \ldots, B_M) \in \mathbb{R}^M$. There are N different types of customers, and for each type $j \in [N]$ there is a permissible set of actions S_j. Each action $s \in S_j$ requires $a_{is} \geq 0$ units of resource i for all $i \in [M]$, which must be available in order for s to be chosen. Choosing action s consumes those resources and earns reward $r_s \geq 0$. We assume that for all types j, there is a "rejection" action $s_{\emptyset j}$ in S_j which does not consume any resources and has no reward. Note that we set a specific rejection action for each S_j. Let $S \triangleq \bigcup_{j \in [N]} S_j$ denote the union of all action sets, and we assume that any two different action sets have no intersection. Let $A = (a_{is})_{M \times |S|} \in \mathbb{R}^{M \times |S|}$ represent the associated consumption matrix, and $R = (r_s)_{|S|} \in \mathbb{R}^{|S|}$ represent the associated reward vector. In particular, $a_{is_{\emptyset j}} = 0$ for all $i \in [M]$ and $r_{s_{\emptyset j}} = 0$ for all $j \in [N]$. We assume $a_{\max} \triangleq \max_{i \in [M], s \in S} a_{is}$ and $r_{\max} \triangleq \max_{s \in S} r_s$. Without loss of generality, we normalize $a_{\min} \triangleq \min_{i \in [M], s \in S: a_{is} > 0} a_{is} = 1$. There are T periods in total. At each time $t = T, T-1, \ldots, 1$, a customer of type j arrives with probability λ_j. No arrival is possible, with probability $1 - \sum_{j \in [N]} \lambda_j$. We assume arrivals are stationary and independent over time (i.e., IID). Define $\lambda_{\min} \triangleq \min_{j \in [N]} \lambda_j$ and $\lambda_{\max} \triangleq \max_{j \in [N]} \lambda_j$. An instance of the online resource allocation problem can be defined as $\mathcal{I} \triangleq \langle B \in \mathbb{R}^M, \{S_j\}_{j \in [N]}, A \in \mathbb{R}^{M \times |S|}, R \in \mathbb{R}^{|S|}, \{\lambda_j\}_{j \in [N]}, T \rangle$. For a given sample path, the realized arrival sequence is denoted as $\Theta \triangleq (\Theta_T, \ldots, \Theta_1)$ with $\Theta_t \in [N]$ for all t. We use $D_j[t_1, t_2] \triangleq \sum_{t=t_2}^{t_1} \mathbb{I}\{\Theta_t = j\}$ to denote the realized demand of customer type

j from time period t_1 to t_2 with $t_1 \geq t_2$. The associated expected demand is $\lambda_j[t_1, t_2] \triangleq \mathbb{E}[D_j[t_1, t_2]] = \lambda_j(t_1 - t_2 + 1)$. For each arriving customer, the decision-maker must decide whether to accept or reject; if accepted, which action to use. The goal is to maximize the total expected reward in the finite horizon. Note that decisions are irrevocable and must be made upon the arrivals of customers.

The aforementioned resource allocation model is general and covers a wide range of classical online problems such as multisecretary (there is only one resource type), online packing or quantity-based network revenue management (each customer type is associated with one action that consumes possibly multiple resources), and online matching (each customer type can be satisfied by using one resource from a given set of options) as described in Vera and Banerjee (2021). In addition, it covers emerging problems in the context of online order fulfillment. For example, each arriving order type j is associated with not only the destination information but also a deadline (e.g., two-day standard shipping, same-day delivery) and S_j is the feasible set of warehouses that can meet the deadline. As another example, for e-retailers with in-house logistics (e.g., Amazon and JD.com), they usually run routine truck schedules between warehouse i and destination j so that there is a truck capacity constraint on the amount of orders that can be transported from warehouse i to destination j. In this case, the resources include both warehouse inventory and truck capacities between warehouses and destinations. The model also covers online order fulfillment in the context of omnichannel. In this setting, the resources are inventory from both warehouses and stores. There are two demand streams: online and offline customers. Offline customers can only be fulfilled by local stores and online customers can be fulfilled by either warehouses or stores.

3.1. Online Resource Allocation with Delay

Instead of making decisions right upon each arrival, now the decision for each arrival can be delayed for K periods. It is more convenient to consider the following equivalent setup: the decision for the arrival Θ_t at time t is still made at time t but with the knowledge of arrivals $\Theta_t, \Theta_{t-1}, \ldots, \Theta_{t-K}$. More specifically, the decision-maker takes an action $s \in S_{\Theta_t}$ based on the realized demand $D_j[t, t-K]$ from time t to $t-K$ and the expected demand $\lambda_j[t-K-1, 1]$ afterward. The classical setting without delay is then captured by setting $K = 0$. In addition, decisions for customers arriving in the last $K+1$ periods $t = K+1, \ldots, 1$ can be made as an offline problem.

To evaluate the performance of online policies with delay, we compare with an optimal offline policy that knows the demand realizations $\Theta = (\Theta_T, \ldots, \Theta_1)$ in advance. The optimal offline can be viewed as a policy with delay $K = T - 1$. Furthermore, we relax the optimal offline to a linear program (LP) that can take an action s any fractional number of times $x_s \in \mathbb{R}$. Formally, for any instance $\mathcal{I}$, any demand vector $\theta = (\theta_{|\theta|}, \ldots, \theta_1)$ and remaining budget vector $b = (b_1, \ldots, b_M)$, define

$$
\begin{aligned}
V^{\text{off}}(\mathcal{I}, \theta, b) \triangleq \ \max \quad & \sum_{s \in S} r_s x_s && (1)\\
\text{s.t.} \quad & \sum_{s \in S} a_{is} x_s \le b_i, && \forall i \in [M],\\
& \sum_{s \in S_j} x_s = \sum_{t=1}^{|\theta|} \mathbb{I}\{\theta_t = j\}, && \forall j \in [N], \qquad (2)\\
& x_s \ge 0, && \forall s \in S : (a_{1s}, \ldots, a_{Ms}) \le b.
\end{aligned}
$$

Variable x_s should be interpreted as the number of times action s is taken, and recall that each s is specific to a type j with S_j including a rejection action $s_{\emptyset j}$, which allows us to impose equality in (2). The optimal offline is defined to be $\mathbb{E}_\Theta[V^{\text{off}}(\mathcal{I}, \Theta, B)]$, where we note that we disallow the optimal offline from taking any infeasible actions s in which $a_{is} > b_i$ for some resource i.

As is standard in the literature, we evaluate the additive loss of an online policy compared to the optimal offline. Let $V^{\text{on}}(\mathcal{I})$ denote the expected reward of the online policy given instance $\mathcal{I}$, and we define *regret* as

$$
\text{REG}_0 \triangleq \ \mathbb{E}_\Theta[V^{\text{off}}(\mathcal{I}, \Theta, B)] - V^{\text{on}}(\mathcal{I}). \qquad (3)
$$

A recent stream of literature (Arlotto and Gurvich 2019, Bumpensanti and Wang 2020, Vera and Banerjee 2021) has shown the surprising result that (even without delay) it is possible to achieve a constant regret, which is not growing with the horizon length T or the initial budget vector B. Our goal is to investigate how this constant regret decreases with delay. We emphasize that it is important that one does not further relax it to the fluid LP, in which the right-hand side of (2) is replaced by the expected demand. The aforementioned literature has shown that it is not possible to achieve constant regret compared to this stronger fluid benchmark.

We now present a general upper bound on REG_0, which is similar to Vera and Banerjee (2021, Lemma 1) but tailored to our setting with delay. For each t, let B_t denote the vector of remaining budgets at time t, and s_t^{on} denote the action taken by the online policy at time t. Note that $B_T = B$.

LEMMA 1. *For any instance $\mathcal{I}$ and online policy, REG_0 is upper bounded by*

$$a_{\max} \cdot M \cdot r_{\max} \cdot \sum_{t=K+2}^{T} \mathbb{P}\left(V^{off}(\mathcal{I}, (\Theta_t, \ldots, \Theta_1), B_t) - V^{off}(\mathcal{I}, (\Theta_{t-1}, \ldots, \Theta_1), B_{t-1}) > r_{s_t^{on}}\right) + \sum_{s \in S} r_s. \tag{4}$$

In words, Lemma 1 is considering the probability of "making a mistake" at each time t before step $K+1$, defined as the *change* in V^{off} *strictly exceeding* the reward collected by the online policy $r_{s_t^{\text{on}}}$. When a mistake is made, the maximal regret that can be incurred comes from the worst case where $a_{\max}$ units of each of the M resources were incorrectly consumed for a low reward, in a world where those resources could have been later consumed to generate a reward of $r_{\max}$ per unit (recall that we normalized $a_{\min}$ to 1). To the best of our knowledge, the analysis of general online allocation problems in Vera and Banerjee (2021, Theorem 5) does not account for this worst-case scenario. The final term in (4) bounds the regret over the last $K+1$ time steps, during which the online policy has full knowledge of demand, and an optimal decision can be obtained by solving an integer program (IP). The integrality gap in the LP for V^{off} could still be as large as $\sum_{s \in S} r_s$, explaining the final term in (4) of Lemma 1, which is formally proven in the appendix.

To investigate the informational benefits of K-period delay, we focus on the regret incurred by an online policy from period T to period $K+2$. For $t = T, ..., K+2$, the decision for Θ_t is based on the actual realizations of $\Theta_t, ..., \Theta_{t-K}$ and forecasts of $\Theta_{t-K-1}, ..., \Theta_1$, in which $\Theta_{t-1}, ..., \Theta_{t-K}$ are K additional periods of information that were unavailable in the no-delay setting. When t reaches $K+1$, decisions for the remaining $K+1$ arrivals are offline, and there is no need to employ any online policy; that is, our proposed algorithm chooses to solve an offline IP in period $t = K+1$, suggesting that the regret from period $K+1$ to period 1 is the integrality gap between two offline optima LP and IP. Therefore, we redefine the regret as the first term in (4) hereafter:

$$\text{REG} \triangleq a_{\max} \cdot M \cdot r_{\max} \cdot \sum_{t=K+2}^{T} \mathbb{P}\left(V^{\text{off}}(\mathcal{I}, (\Theta_t, \ldots, \Theta_1), B_t) - V^{\text{off}}(\mathcal{I}, (\Theta_{t-1}, \ldots, \Theta_1), B_{t-1}) > r_{s_t^{\text{on}}}\right). \tag{5}$$

Note that for some special problems, such as the multisecretary problem in Section 4, the integrality gap is zero. This implies that the partial regret defined in (5) is exactly equal to the total regret.

In the following sections, we will show the regret REG of the online allocation problem decays exponentially w.r.t. K. In addition to having completely offline decisions that eliminate the regret in periods $t = K+1, \ldots, 1$, this result comes from the fact that the regret in each individual period $t = T, \ldots, K+2$ also exhibits exponential decay w.r.t. K, benefiting from the additional information provided by the K-period delay.

4. Multisecretary Problem

Before proceeding to the general online resource allocation problem as described in Section 3, in this section we focus on a simpler case — the multisecretary problem and analyze the benefits of delay. Let us first briefly review the multisecretary problem. There are available secretary positions and the decision-maker must decide whether to accept or reject each arriving candidate to maximize the total capability of accepted candidates. This is a special case of the general resource allocation problem as described in Section 3. Specifically, there is only one type of resource (i.e., $M = 1$) with initial budget $B \in \mathbb{N}$ (i.e., B positions). For each candidate type $j \in [N]$, the action set is the same: either accept or reject. Acceptance requires one unit of resource and the reward is r_j. Without loss of generality, let $0 < r_1 < r_2 < \cdots < r_N$. There is one and only one candidate arriving at each time period, namely, $\sum_{j\in[N]} \lambda_j = 1$. Let $F_j \triangleq \lambda_1 + \lambda_2 + \cdots + \lambda_j$ be the cumulative distribution function (CDF) of the arrival distribution. There are T candidates in total. Let $x_j, x_{\emptyset j}$ denote the number of accepted and rejected type j candidates, respectively. We use $B_t \in \{0, 1, 2, \ldots, B\}$ to denote the remaining budget when making decisions for Θ_t.

We next define two LPs. The offline LP for the multisecretary problem can be formulated as:

$$
\begin{aligned}
\max \quad & \sum_{j\in[N]} r_j x_j && (6)\\
\text{s.t.} \quad & \sum_{j\in[N]} x_j \le B_t, \\
& x_j + x_{\emptyset j} = D_j[t,1], && \forall j \in [N], \\
& x_j,\ x_{\emptyset j} \ge 0, && \forall j \in [N].
\end{aligned}
$$

With a $K-$period delay, the decision-maker has information about the realized demand $D_j[t, t-K]$ and expected demand $\lambda_j[t-K-1, 1]$ at time period t, and the online LP for the multisecretary problem can be formulated as follows:

$$\begin{aligned} \max \quad & \sum_{j\in[N]} r_j x_j & & (7) \\ \text{s.t.} \quad & \sum_{j\in[N]} x_j \le B_t, & & \\ & x_j + x_{\emptyset j} = D_j[t, t-K] + \lambda_j[t-K-1, 1], & \forall j \in [N], & \\ & x_j,\ x_{\emptyset j} \ge 0, & \forall j \in [N]. & \end{aligned}$$

Let $\left\{X^*_{t,j},\ X^*_{t,\emptyset j}\right\}_{j\in[N]}$ and $\{X_{t,j},\ X_{t,\emptyset j}\}_{j\in[N]}$ be the optimal solutions to the offline and online LP Problems (6) and (7), respectively. Many well-performing online algorithms in the literature are constructed by (re-)solving the online LP. Our online algorithm (to be described in Algorithm 1) is no exception and resolves the online LP Problem (7) for each t whenever the state updates. The offline LP Problem (6) is also defined for each t and serves as a benchmark for us to evaluate how good is the solution derived from the online LP Problem (7). Note that the only difference between LPs (6) and (7) is on the right-hand side: the offline LP sees all realized demands $D_j[t, 1]$ from time t to 1, whereas the online LP with delay only sees realized demands $D_j[t, t-K]$ from time t to $t-K$ and later demands $\lambda_j[t-K-1, 1]$ from time $t-K-1$ to 1 are only known in expectation. The algorithm is formally stated in Algorithm 1.

Following the idea of Vera and Banerjee (2021), Algorithm 1 "correctly" balances the probabilities of an incorrect acceptance vs. rejection, and accepts the candidate with type Θ_t at time t if and only if (iff) $X_{t,\Theta_t} \ge X_{t,\emptyset\Theta_t}$. Equivalently, the algorithm ends up accepting a type j iff

$$B_t - \sum_{\ell=j+1}^{N} \left(D_\ell[t, t-K] + \lambda_\ell[t-K-1, 1]\right) \ge \frac{D_j[t, t-K] + \lambda_j[t-K-1, 1]}{2}, \tag{8}$$

(we defer the derivation of (8) to the appendix). In the classical setting where $K = 0$ (no delay), conditional on $\Theta_t = j$, acceptance criterion (8) can be simplified to

$$B_t - \sum_{\ell=j+1}^{N} \lambda_\ell[t-1, 1] \ge \frac{1 + \lambda_j[t-1, 1]}{2}, \tag{9}$$

Algorithm 1: Delayed Bayesian Prophet for Multisecretary Problem

```
for t = T, T-1, ..., K+2 do
    Observe Θ_t, Θ_{t-1}, ..., Θ_{t-K};
    if B_t ≥ (1 - F_{Θ_t} + λ_{Θ_t}/2)(t-K-1) + ( Σ_{ℓ=Θ_t+1}^{N} D_ℓ[t,t-K] + (1/2)·D_{Θ_t}[t,t-K] ) then
        Accept Θ_t and update B_{t-1} ← B_t - 1;
    else
        Reject Θ_t and let B_{t-1} ← B_t;
    end
end
Observe Θ_{K+1}, ..., Θ_1 and accept B_{K+1} candidates with highest r_{Θ_t} for t = K+1, ..., 1.
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because $D_\ell[t,t-K] = D_\ell[t,t] = 0$ unless $\ell = j$ (in which case it equals 1). This is an improvement of Vera and Banerjee (2021), whose acceptance criterion is equivalent to

$$B_t - \sum_{\ell=j+1}^{N} \lambda_\ell[t,1] \geq \frac{\lambda_j[t,1]}{2}. \tag{10}$$

Their algorithm is essentially deciding the acceptance threshold *before* realizing the type of Θ_t. The improved algorithm implied by our delay framework makes better use of available information, and has a regret bound smaller by a factor of ρ_0 (see Remark 1 below). For notational convenience, we refer to (10) as the setting with $K = -1$.

Intuitively, the online LP Problem (7) becomes closer to the offline LP Problem (6) as K increases, and it is interesting to analyze Algorithm 1's performance and convergence rate in K. We present an exponential bound in Theorem 1. Define $\rho_0 \triangleq \lambda_{\min} e^{-\lambda_{\min}} + (1-\lambda_{\min}) < 1$. Note that $a_{\max} = 1$, $M = 1$, and $r_{\max} = r_N$.

THEOREM 1. *For any instance $\mathcal{I}$ and $K \geq 0$, the expected regret of Algorithm 1 has the following upper bound:*

$$REG \leq \rho_0^K \cdot \frac{2 \cdot r_N \cdot e^{-\lambda_{\min}}}{\lambda_{\min}^2}. \tag{11}$$

The proof sketch is provided below, with the full version of the proof deferred to the appendix. We first present necessary conditions for incorrect acceptance and rejection. Note that incorrect decisions occur when the solution of the online LP Problem (7) deviates from that of the offline LP Problem (6). As discussed earlier, the gap between the offline and online LPs arises from the right-hand side: one uses the realized demand $D_j[t,1]$, while

the other uses the realized demand $D_j[t, t-K]$ plus the expected demand $\lambda_j[t-K-1, 1]$. More specifically, one can verify that incorrect acceptance for Θ_t occurs iff $X^*_{t,\Theta_t} = 0$ and (8) holds, and incorrect rejection for Θ_t occurs iff $X^*_{t,\emptyset\Theta_t} = 0$ and the reverse of (8) holds. Note that

$$X^*_{t,\Theta_t} = 0 \iff \sum_{\ell=\Theta_t+1}^{N} D_\ell[t,1] \geq B_t, \text{ and } X^*_{t,\emptyset\Theta_t} = 0 \iff \sum_{\ell=\Theta_t}^{N} D_\ell[t,1] \leq B_t. \tag{12}$$

Combining (12) and (8), it follows that incorrect acceptance implies

$$\sum_{\ell=\Theta_t+1}^{N} D_\ell[t-K-1,1] - (1-F_{\Theta_t})(t-K-1) \geq \frac{\lambda_{\Theta_t}}{2} \cdot (t-K-1) + \frac{1}{2} \cdot D_{\Theta_t}[t, t-K], \tag{13}$$

and incorrect rejection implies

$$(1 - F_{\Theta_t} + \lambda_{\Theta_t})(t-K-1) - \sum_{\ell=\Theta_t}^{N} D_\ell[t-K-1,1] \geq \frac{\lambda_{\Theta_t}}{2} \cdot (t-K-1) + \frac{1}{2} \cdot D_{\Theta_t}[t, t-K]. \tag{14}$$

The left-hand sides of (13) and (14) correspond to the difference between the right-hand sides of the offline and online LPs.

Next, from Hoeffding's inequality, the probabilities of events (13) and (14) are both at most

$$\exp\left(-\frac{t-K-1}{2} \cdot \left(\lambda_{\Theta_t} + \frac{D_{\Theta_t}[t, t-K]}{t-K-1}\right)^2\right).$$

Thus, conditional on Θ_t and taking the expectation over $(\Theta_{t-1}, \ldots, \Theta_{t-K})$, one can derive

$$\begin{aligned}
&\mathbb{P}\left(V^{\text{off}}(\mathcal{I}, (\Theta_t, \ldots, \Theta_1), B_t) - V^{\text{off}}(\mathcal{I}, (\Theta_{t-1}, \ldots, \Theta_1), B_{t-1}) > r_{s_t^{\text{on}}}\right) \\
\leq\ & \mathbb{E}\left[\exp\left(-\frac{t-K-1}{2} \cdot \left(\lambda_{\Theta_t} + \frac{D_{\Theta_t}[t, t-K]}{t-K-1}\right)^2\right) \middle| \Theta_t\right] \\
\leq\ & \exp\left(-\frac{\lambda_{\Theta_t}^2}{2} \cdot (t-K-1)\right) \cdot e^{-\lambda_{\Theta_t}} \cdot \left(\lambda_{\Theta_t} e^{-\lambda_{\Theta_t}} + (1-\lambda_{\Theta_t})\right)^K.
\end{aligned} \tag{15}$$

Finally, with $a_{\max} = M = 1$ and $r_{\max} = r_N$, the regret in (5) reduces to

$$\text{REG} \leq r_N \cdot \sum_{t=K+2}^{T} \mathbb{P}\left(V^{\text{off}}(\mathcal{I}, (\Theta_t, \ldots, \Theta_1), B_t) - V^{\text{off}}(\mathcal{I}, (\Theta_{t-1}, \ldots, \Theta_1), B_{t-1}) > r_{s_t^{\text{on}}}\right). \tag{16}$$

Combining (16) and (15) completes the proof of Theorem 1.

Remark 1. When there is no delay (i.e., $K=0$), our bound in (11) reduces to $\frac{2\cdot r_N\cdot e^{-\lambda_{\min}}}{\lambda_{\min}^2}$, which improves upon the guarantee of $\frac{2\cdot r_N}{\lambda_{\min}^2}$ implied by Vera and Banerjee (2021) by a factor of ρ_0 (note that $e^{-\lambda_{\min}}\le\rho_0$), because our improved algorithm makes better use of available information as discussed aforementioned.

Theorem 1 shows that for a fixed K, the regret for the multisecretary problem of Algorithm 1 is upper bounded by a constant independent of the horizon length T and the budget B, which is consistent with results in Vera and Banerjee (2021). As mentioned before, there is no integrality gap between the offline LP (6) and its IP version, and the regret bound for the total T periods is exactly (5). Thus, Theorem 1 implies that the total regret bound decays exponentially fast in the length of the delay K, with the base ρ_0 depending only on the underlying probability distribution. Hence, a little delay can significantly reduce the regret, regardless of T and B. Furthermore, from the proof sketch, we can observe that the probability of making a mistake at any particular period $t=T,\dots,K+2$ also has an upper bound that decays exponentially. More specifically, our analysis suggests that the probability of making a mistake decreases with D_{Θ_t} (the number of arrivals of the same type appearing in the next K periods), and (15) establishes the following upper bound for the regret in period t with a K-period delay, denoted as $\mathrm{RB}(t,K)$:

$$\mathrm{RB}(t,K)\triangleq\ r_N\cdot e^{-\lambda_{\min}}\cdot\left(\lambda_{\min}e^{-\lambda_{\min}}+(1-\lambda_{\min})\right)^K\cdot\exp\left(-\frac{\lambda_{\min}^2}{2}\cdot(t-K-1)\right). \tag{17}$$

Note that $\mathrm{RB}(t,0)=r_N\cdot e^{-\lambda_{\min}}\cdot\exp\left(-\frac{\lambda_{\min}^2}{2}\cdot(t-1)\right)$ and

$$\frac{\mathrm{RB}(t,K)}{\mathrm{RB}(t,0)}=\left(\frac{\lambda_{\min}e^{-\lambda_{\min}}+(1-\lambda_{\min})}{\exp\left(-\lambda_{\min}^2/2\right)}\right)^K\ <\ 1, \tag{18}$$

because $xe^{-x}+(1-x)<\exp\left(-x^2/2\right)$ for $x>0$. Thus, (18) shows that the ratio $\frac{\mathrm{RB}(t,K)}{\mathrm{RB}(t,0)}$ also decays exponentially in K, which demonstrates the informational benefits of a K-period lookahead for reducing the regret at any single period t.

Remark 2. It is possible to achieve exponential decay by excluding the last $(K+1)$ periods and applying existing regret bounds from Vera and Banerjee (2021). More specifically, consider an algorithm that does not delay during periods $t=T,\dots,K+2$, and employs a K-period delay starting at period $t=K+1$, incurring zero regret from that point onward. This algorithm would exhibit regret that decays exponentially in K; however, even using our improved bound (17), the base for the exponential decay would be

$\exp(-\lambda_{\min}^2/2)$, which is slower than the rate $\rho_0 = \lambda_{\min} e^{-\lambda_{\min}} + (1-\lambda_{\min})$ established in Theorem 1. Furthermore, this algorithm would not achieve the improved regret for periods before $K+1$ in the way our "delayed Bayesian prophet" algorithm achieves (18) for all periods t.

5. General Online Resource Allocation Problem

In this section, we extend our study to the general online resource allocation problem with delay as described in Section 3. Similar to the multisecretary problem, our online policy solves an online LP with updated state and demand information for each arriving customer and the allocation decision is based on the optimal solution of the online LP. However, even with $K=0$, if we naively adopt the Bayes selector for online allocation in Vera and Banerjee (2021, Algorithm 4) (i.e., by solving an online LP with a fixed coefficient matrix for updated recourse amounts and demand expectations in each period), it may lead to infeasible allocations. The infeasibility comes from the multi-unit consumption $a_{is} \in \mathbb{N}$, which can be illustrated through the following example.

Example 1. There is a single resource i with $B_i = 2$, a single customer type j with a single non-reject action s with $a_{is} = 3$. At the beginning of period t, type j does not arrive and $\lambda_j[t-1,1] = 1$. Then, $S_j = \{s, s_\emptyset\}$ and the optimal solution of the LP is $x_{sj} = 2/3$ and $x_{s_\emptyset j} = 1/3$. Algorithm 4 in Vera and Banerjee (2021) would suggest the non-reject action s for j. However, this action is infeasible due to the insufficient amount of resource i. Thus, we must reject j.

This type of rejection due to infeasible actions as illustrated via Example 1 means that the algorithm may not be able to take the action suggested by Vera and Banerjee (2021, Algorithm 4). To address this infeasibility issue, we modify the policy by adding an additional step to update the action sets. Specifically, before re-solving the LP each time, we first check the feasibility of all actions and remove infeasible ones: namely, given the current budget vector B_t, for each action $s \in S_j$, if $a_{is} \le B_{t,i}$ for all $i \in [M]$, then s is feasible and remains in the action set; otherwise, it is removed. This modification ensures that all allocations made by the online LP are feasible and there is no rejection due to infeasibility.

We emphasize that we are still using the overall Bayesian prophet framework from Vera and Banerjee (2021), and that this modification is unnecessary for $a_{is} \in \{0,1\}$ (e.g.,

online matching) because this infeasibility issue can never happen. Generally, we denote the updated feasible action sets at time t by $\{S_{t,j}\}_{j\in[N]}$. Without loss of generality, we assume that the initial action sets are feasible; namely, $S_{T,j} = S_j$ for all $j \in [N]$. We further define $S_t \triangleq \bigcup_{j\in[N]} S_{t,j}$.

Similar to the multisecretary problem, we next define two LPs. The offline LP (accounting for this modification) can be formulated as follows:

$$\begin{aligned} \max \quad & \sum_{s\in S_t} r_s x_s & & (19)\\ \text{s.t.} \quad & \sum_{s\in S_t} a_{is} x_s \le B_{t,i}, & \forall i \in [M],\\ & \sum_{s\in S_{t,j}} x_s = D_j[t,1], & \forall j \in [N],\\ & x_s \ge 0, & \forall s \in S_t. \end{aligned}$$

The online LP (accounting for both this modification and the ability to delay) can be formulated as follows:

$$\begin{aligned} \max \quad & \sum_{s\in S_t} r_s x_s & & (20)\\ \text{s.t.} \quad & \sum_{s\in S_t} a_{is} x_s \le B_{t,i}, & \forall i \in [M],\\ & \sum_{s\in S_{t,j}} x_s = D_j[t,t-K] + \lambda_j[t-K-1,1], & \forall j \in [N],\\ & x_s \ge 0, & \forall s \in S_t. \end{aligned}$$

Let $\{X^*_{t,s}\}_{s\in S_t}$ and $\{X_{t,s}\}_{s\in S_t}$ be an optimal solution to the offline LP Problem (19) and online LP Problem (20), respectively (note that these two problems coincide for the last $K+1$ arrivals $\Theta_{K+1}, \Theta_K, \ldots, \Theta_1$). The algorithm is then formally stated in Algorithm 2.

From Algorithm 2, it is straightforward to verify that the cardinality of the action set $S_{t,j}$ is non-increasing in t for each j (i.e., $S_{t,j}$ shrinks over the time) and $X_{t,s_t^{\text{on}}} = \max_{s\in S_{t,\Theta_t}} \{X_{t,s}\}$ implies

$$X_{t,s_t^{\text{on}}} \ge \frac{D_{\Theta_t}[t,t-K] + \lambda_{\Theta_t}[t-K-1,1]}{C_{\max}+1}, \tag{21}$$

where $C_{\max} \triangleq \max_{j\in[N]} \{|S_{T,j} \setminus s_{\emptyset j}|\} = \max_{j\in[N]} \{|S_j \setminus s_{\emptyset j}|\}$. Note that (21) is critical for the regret analysis. The right-hand side of (21) also provides clues to design a family of online policies

Algorithm 2: Delayed Bayesian Prophet for General Online Resource Allocation Problem

for $t = T, T-1, \ldots, K+2$ **do**
  Update the feasible action sets $\{S_{t,j}\}_{j\in[N]}$;
  Observe $\Theta_t, \Theta_{t-1}, \ldots, \Theta_{t-K}$;
  Solve Problem (20) and obtain an optimal solution $\{X_{t,s}\}_{s\in S_t}$;
  Choose action $s_t^{\text{on}} \leftarrow \arg\max_{s\in S_{t,\Theta_t}} \{X_{t,s}\}$, breaking ties arbitrarily;
  Update $B_{t-1,i} \leftarrow B_{t,i} - a_{is_t^{\text{on}}}$ for each $i \in [M]$;
end
Update $\{S_{K+1,j}\}_{j\in[N]}$ and observe $\Theta_{K+1}, \Theta_K, \ldots, \Theta_1$; follow the offline optimal hindsight policy for the last $K+1$ allocations.

that can lead to the same order of regret as the current policy, and in fact we can choose any action $s_t^{\text{on}} \in S_{t,\Theta_t}$ satisfying $X_{t,s_t^{\text{on}}} \geq \frac{D_{\Theta_t}[t,t-K]+\lambda_{\Theta_t}[t-K-1,1]}{C_{\max}+1}$.

The following sensitivity result (or Lipschitz property) of optimal solutions on the right-hand side demand vector in the general online allocation problem is critical for our regret analysis. Following Vera and Banerjee (2021), we make use of Theorem 2.4 in Mangasarian and Shiau (1987), but we need to specifically define the Lipschitz constant due to our action set S_t that shrinks over time t. To elaborate, let J_t denote the constraint matrix of Problems (19) and (20) at time t, including all of the inequality and equality constraints. Note that $J_t \in \mathbb{R}^{(M+N+|S_t|)\times|S_t|}$. Mangasarian and Shiau (1987) show for some constant $\kappa_t \triangleq \kappa(J_t)$ depending only on matrix J_t that: given any optimal solution X_t to the online Problem (20), there exists an optimal solution X_t^* to the offline Problem (19) such that

$$\begin{aligned} \|X_t^* - X_t\|_\infty &\leq \kappa_t \|D[t,1] - (D[t,t-K] + \lambda[t-K-1,1])\|_\infty \\ &= \kappa_t \|D[t-K-1,1] - \lambda[t-K-1,1]\|_\infty . \end{aligned} \tag{22}$$

We claim that the constant bound κ_t is non-increasing t, which allows us to define $\kappa_{\max} \triangleq \kappa_T \geq \kappa_t$ for $t = T, T-1, \ldots, K+2$.

To prove $\kappa_{t+1} \geq \kappa_t$, note that the exact specification of κ_t in Theorem 2.4 of Mangasarian and Shiau (1987) can be re-written as

$$\kappa_t = \sup_{u\in\mathbb{R}^{M+N+|S_t|}} \left\{ \|u\|_1 \middle| \|u' J_t\|_1 = 1, \text{the rows of } J_t \text{ with nonzero entries of } u \text{ are linearly independent} \right\}. \tag{23}$$

Note that matrix J_t is equal to matrix J_{t+1} but with the rows and columns corresponding to actions s that became infeasible at time t removed. Therefore, given any feasible u in Problem (23) for J_t, we can define a feasible u in Problem (23) for J_{t+1} by setting the entries of u for actions not feasible at t to 0, thereby showing that $\kappa_{t+1} \geq \kappa_t$.

Next, we state the main result of the paper: an upper bound on the expected regret of Algorithm 2. Recall that $a_{\max} = \max_{i\in[M], s\in S} a_{is}$, $r_{\max} = \max_{s\in S} r_s$, $\lambda_{\min} = \min_{j\in[N]} \lambda_j$, $\lambda_{\max} = \max_{j\in[N]} \lambda_j$, and $C_{\max} = \max_{j\in[N]} \{|S_j \setminus s_{\emptyset j}|\}$. We additionally define $C_1 \triangleq (a_{\max}+1)C_{\max} + a_{\max}$,

$$\rho_1 \triangleq \exp(-2\lambda_{\min}^2), \quad \rho_2 \triangleq \lambda_{\min} \exp\left(-\frac{4\lambda_{\min}}{\kappa_{\max}^2 (C_{\max}+1)^2}\right) + (1-\lambda_{\min}), \quad \rho \triangleq \max\{\rho_1, \rho_2\} < 1.$$

THEOREM 2. *For any instance $\mathcal{I}$ and $K \geq 0$, the expected regret of Algorithm 2 has the following upper bound:*

$$a_{\max} \cdot M \cdot r_{\max} \cdot \left(\left\lceil \frac{C_1}{\lambda_{\min}} \right\rceil \cdot \rho_1^{-2\frac{C_1}{\lambda_{\min}}} + \frac{\kappa_{\max}^2 \cdot (C_{\max}+1)^2}{\lambda_{\min}^2} \cdot \exp\left(\frac{4\lambda_{\max} \cdot C_1}{\kappa_{\max}^2 \cdot (C_{\max}+1)^2} \right) \right) \cdot \rho^K. \tag{24}$$

Theorem 2 shows that the upper bound on expected regret under the policy decays exponentially in the length of the delay K. The upper bound in (24) is independent of T and B. Although $\{S_{t,j}\}_{j\in[N]}$ and the constraint matrix J_t in Problem (20) can be updated over time, the base ρ and the coefficient depend only on the initial parameters $\{S_{T,j}\}_{j\in[N]}$ and J_T. In particular, for a special case - the multisecretary problem - the bound is worse than the one in (11). This is because we use $X^*_{t,s_t^{\text{on}}} < a_{\max}+1$ as the necessary condition for incorrect decisions to derive (24) (see more details in the proof of Theorem 2), whereas this condition can be greatly simplified to $X^*_{t,s_t^{\text{on}}} = 0$ in the special case.

6. Extensions

In this section, we consider three extensions: (i) the setting where the arrival distribution is independent but non-identical; (ii) the setting where the arrival distribution is unknown; and (iii) the setting where arrivals are cleared in a batch every $K+1$ periods, rather than each individual arrival being delayed for K periods on a rolling basis. We also compare the results of these two settings with those of the baseline setting studied in Section 5.

6.1. Independent but Non-Identical Arrival Distribution

In this section, we study a general online allocation problem with independent but non-identical arrival distributions. Let $\lambda_{t,j}$ denote the arrival probability of type j in period

t, and define $\bar{\lambda}_{\min} \triangleq \min_{t\in[T],j\in[N]} \lambda_{t,j}$. The expected demand of type j from period t_1 to t_2, inclusive, is $\lambda_j[t_1,t_2] = \sum_{t=t_2}^{t_1} \lambda_{t,j}$. We assume $\bar{\lambda}_{\min} > 0$; that is, for each customer type j, there is a positive probability of arrival in each period t. Note that this assumption implies $\lambda_j[t,1] > 0$ for each $j \in [N]$ and $t \in [T]$.

It is straightforward to see that (20) and Algorithm 2 remain valid after the corresponding modifications in this general setting. Based on the analysis in Section 5 and applying Hoeffding's inequality for independent random variables, we derive the following corollary from (24). Define $\bar{\lambda}_{\max} = \max_{t\in[T],j\in[N]} \lambda_{t,j}$, and recall that $a_{\max} = \max_{i\in[M],s\in S} a_{is}$, $r_{\max} = \max_{s\in S} r_s$, $C_{\max} = \max_{j\in[N]} \{|S_j \setminus s_{\emptyset j}|\}$, $C_1 = (a_{\max}+1)C_{\max} + a_{\max}$ and $\kappa_{\max} = \kappa_T$. We additionally define

$$\bar{\rho}_1 \triangleq \exp(-2\bar{\lambda}_{\min}^2), \quad \bar{\rho}_2 \triangleq \bar{\lambda}_{\min} \exp\left(-\frac{4\bar{\lambda}_{\min}}{\kappa_{\max}^2 (C_{\max}+1)^2}\right) + (1-\bar{\lambda}_{\min}), \quad \bar{\rho} \triangleq \max\{\bar{\rho}_1, \bar{\rho}_2\} < 1.$$

Corollary 1. *For any instance $\mathcal{I}$ with $\bar{\lambda}_{\min} > 0$ and $K \geq 0$, the expected regret of Algorithm 2 has the following upper bound:*

$$a_{\max} \cdot M \cdot r_{\max} \cdot \left(\left\lceil \frac{C_1}{\bar{\lambda}_{\min}} \right\rceil \cdot \bar{\rho}_1^{-2\frac{C_1}{\bar{\lambda}_{\min}}} + \frac{\kappa_{\max}^2 \cdot (C_{\max}+1)^2}{\bar{\lambda}_{\min}^2} \cdot \exp\left(\frac{4\bar{\lambda}_{\max} \cdot C_1}{\kappa_{\max}^2 \cdot (C_{\max}+1)^2} \right) \right) \cdot \bar{\rho}^K.$$

Corollary 1 extends these informational benefits of delay to the non-IID setting. As previously discussed in (18), the information revealed by a K-period delay allows us to reduce the regret upper bound in the case of no delay by a factor of $\left(\frac{\bar{\lambda}_{\min} e^{-\bar{\lambda}_{\min}} + (1-\bar{\lambda}_{\min})}{\exp(-\bar{\lambda}_{\min}^2/2)} \right)^K$ for each period $t = T, \ldots, K+2$. Intuitively, delay provides greater informational benefits when demand information is evenly distributed across periods. In contrast, consider the case without the assumption $\bar{\lambda}_{\min} > 0$, such as an extreme example where there is a positive probability of customer arrival only at time steps that are more than K periods apart. In this scenario, it is clear that a K-period delay offers no benefit, because the online LP with a K-period delay would be identical to one without any delay.

6.2. Unknown Arrival Distribution

In this section, we study a general online allocation problem in which the decision-maker knows that arrivals are IID but has no knowledge of its distribution. The only difference from the baseline setting as described in Section 5 is that, while the decision-maker knows the arrival follows an IID distribution, they have no knowledge of the true arrival rates $\{\lambda_j\}_{j\in[N]}$. As such, the decision-maker must estimate the true distribution in each period

t by using past realizations. We modify Algorithm 2 to include an additional estimation step in each period by using the sample average, and show that the expected regret of this modified algorithm is still upper bounded by a constant that decays exponentially fast in K.

Let $\hat{\lambda}_{t,j}$ be the estimated arrival rate of type j in period t. It is estimated by using the frequency of type j appearing in the past/current samples $\Theta_T, \ldots, \Theta_t$ as well as the K-period advanced information samples $\Theta_{t-1}, \ldots, \Theta_{t-K}$:

$$\hat{\lambda}_{t,j} = \frac{D_j[T, t-K]}{T-(t-K)+1}.$$

Based on $\hat{\lambda}_{t,j}$, the expected demand of type j from period $t-K-1$ to period 1 is $\hat{\lambda}_{t,j}[t-K-1,1] = \hat{\lambda}_{t,j} \cdot (t-K-1)$. The online LP can be formulated accordingly:

$$\begin{aligned} \max \quad & \sum_{s \in S_t} r_s x_s & & (25) \\ \text{s.t.} \quad & \sum_{s \in S_t} a_{is} x_s \le B_{t,i}, & \forall i \in [M], \\ & \sum_{s \in S_{t,j}} x_s = D_j[t, t-K] + \hat{\lambda}_{t,j}[t-K-1, 1], & \forall j \in [N], \\ & x_s \ge 0, & \forall s \in S_t. \end{aligned}$$

The algorithm, as formally stated in Algorithm 3, is identical to Algorithm 2 except using the estimated arrival rates rather than the true ones.

Although the online decision made in period t depends on the estimation $\{\hat{\lambda}_{t,j}\}_{j \in [N]}$ rather than the true distribution $\{\lambda_j\}_{j \in [N]}$, we can apply Hoeffding's inequality to bound the probability of misestimation, which is exponential in the length of realizations (i.e., $T-(t-K)+1$). Thus, the summation of the probability of incorrect estimation over the T-period horizon is still a constant. We present in Theorem 3 an upper bound on the expected regret of Algorithm 3. Recall that $C_1 = (a_{\max}+1)C_{\max} + a_{\max}$. Define

$$\begin{aligned} &\eta \triangleq \frac{\lambda_{\min}}{3} \cdot \min\left\{\frac{1}{\kappa_{\max}(C_{\max}+1)},\ 1\right\}, \qquad \hat{\rho}_1 \triangleq \exp\left(-2\left(\frac{\lambda_{\min}}{3}\right)^2\right), \\ &\hat{\rho}_2 \triangleq \frac{\lambda_{\min}}{3} \cdot \exp\left(-\frac{4\lambda_{\min}/3}{\kappa_{\max}^2 (C_{\max}+1)^2}\right) + \left(1 - \frac{\lambda_{\min}}{3}\right), \quad \hat{\rho}_3 \triangleq e^{-2\eta^2}, \qquad \hat{\rho} \triangleq \max\{\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3\}. \end{aligned} \tag{26}$$

Algorithm 3: Delayed Bayesian Prophet for General Online Resource Allocation Problem with Unknown Distribution

for $t = T, T-1, \ldots, K+2$ **do**
  Update the estimated arrival distribution by $\hat{\lambda}_{t,j} = \frac{D_j[T,t-K]}{T-(t-K)+1}$ for all $j \in [N]$;
  Update the feasible action sets $\{S_{t,j}\}_{j\in[N]}$;
  Observe $\Theta_t, \Theta_{t-1}, \ldots, \Theta_{t-K}$;
  Solve Problem (25) and obtain an optimal solution $\{X_{t,s}\}_{s\in S_t}$;
  Choose action $s_t^{\text{on}} \leftarrow \arg\max_{s\in S_{t,\Theta_t}} \{X_{t,s}\}$, breaking ties arbitrarily;
  Update $B_{t-1,i} \leftarrow B_{t,i} - a_{is_t^{\text{on}}}$ for each $i \in [M]$;
end
Update $\{S_{K+1,j}\}_{j\in[N]}$ and observe $\Theta_{K+1}, \Theta_K, \ldots, \Theta_1$; follow the offline optimal hindsight policy for the last $K+1$ allocations.

THEOREM 3. *For any instance $\mathcal{I}$ and $K \geq 0$, the expected regret of Algorithm 3 has the following upper bound:*

$$a_{\max} \cdot M \cdot r_{\max} \cdot \left(\frac{2}{\eta^2} + \left\lceil \frac{3C_1}{\lambda_{\min}} \right\rceil \cdot \hat{\rho}_1^{-2\frac{C_1}{\lambda_{\min}/3}} + \frac{\kappa_{\max}^2 \cdot (C_{\max}+1)^2}{(\lambda_{\min}/3)^2} \cdot \exp\left(\frac{4 \cdot (\lambda_{\max}/3) \cdot C_1}{\kappa_{\max}^2 \cdot (C_{\max}+1)^2} \right) \right) \cdot \hat{\rho}^K. \tag{27}$$

Theorem 3 shows that we can still achieve constant regret independent of T even without knowing the distribution, and the upper bound in (27) also decays exponentially in the length of the delay K. Moreover, it is intuitive to conjecture that delay provides additional benefits in this setting due to distribution estimation. An interesting question is whether delay generally results in more benefits in the setting with unknown distribution compared to the setting with known distribution. We conduct numerical experiments in Section 7.2 that suggest an affirmative answer to this question.

6.3. Batching

In this section, we study a general online resource allocation problem with batching. Instead of delaying each decision by K periods, now all decisions are made in batches with size $K+1$ (note that both reduce to the classical case without delay/batching in the special case where $K=0$). We modify Algorithm 2 to incorporate batching instead of delay, calling it "batched Bayesian prophet," and provide an upper bound on the expected regret. We also compare the batching algorithm to the one with delay (i.e., Algorithm 2), and interpret

batching as a form of heterogeneous delay—arrivals in the same batch are delayed by K periods, $K-1$ periods, ..., and 0 periods, respectively.

Let us first formulate the online resource allocation problem with K-period batching. Here, the notation K-period batching means that each batch has $K+1$ arrivals. For any T, we have $T=(K+1)(\lceil T/(K+1)\rceil-1)+R$ for some $R\in[K+1]$. Specifically, there are $\lceil T/(K+1)\rceil$ batches in total, where each of the first $\lceil T/(K+1)\rceil-1$ batches covers $(K+1)$ periods, and the last batch covers the remaining R periods. It suffices to consider the regret incurred in the first $\lceil T/(K+1)\rceil-1$ batches. For $n=1,\ldots,\lceil T/(K+1)\rceil-1$, the n-th batch consists of $\Theta_{T-(n-1)(K+1)},\Theta_{T-(n-1)(K+1)-1},\ldots,\Theta_{T-n(K+1)+1}$, and decisions for them are made based on the knowledge of these realized arrivals in the batch. When processing each batch, our algorithm still commits to decisions sequentially. In particular, we treat a batch as consisting of arrivals whose decisions are made with $m=K,K-1,\ldots,0$ periods of lookahead; namely, the decision for arrival $\Theta_{T-(n-1)(K+1)-(K-m)}$ (in the n-th batch) is made based on realizations $\Theta_{T-(n-1)(K+1)-(K-m)},\ldots,\Theta_{T-n(K+1)+1}$ and the expected demand $\lambda[T-n(K+1),1]$, treating decisions for arrivals before $T-(n-1)(K+1)-(K-m)$ as committed. This is equivalent to sequential processing where arrival $T-(n-1)(K+1)-(K-m)$ has an m-period delay. Note that the last batch consisting of $\Theta_R,\Theta_{R-1},\ldots,\Theta_1$ can be regarded as an offline problem. For any period t, the online LP is equivalent to an online LP with $m(K,t)$-period delay, where we define

$$m(K,t)\triangleq K-\left(T-t-(K+1)\left\lfloor\frac{T-t}{K+1}\right\rfloor\right),\ \forall t=T,T-1,\ldots,R+1.$$

The online LP with K-period batching can then be formulated as follows:

$$\begin{aligned}
\max\quad & \sum_{s\in S_t} r_s x_s && && (28)\\
\text{s.t.}\quad & \sum_{s\in S_t} a_{is}x_s\le B_{t,i}, && \forall i\in[M],\\
& \sum_{s\in S_{t,j}} x_s = D_j[t,t-m(K,t)]+\lambda_j[t-m(K,t)-1,1], && \forall j\in[N],\\
& x_s\ge 0, && \forall s\in S_t.
\end{aligned}$$

We apply the idea of Algorithm 2 to this model and formally state our algorithm in Algorithm 4.

Algorithm 4: Batched Bayesian Prophet for General Online Resource Allocation Problem

for $t = T, T-1, \ldots, R+1$ **do**
  Update the feasible action sets $\{S_{t,j}\}_{j\in[N]}$;
  Observe $\Theta_t, \Theta_{t-1}, \ldots, \Theta_{t-m(K,t)}$;
  Solve Problem (28) and obtain an optimal solution $\{X_{t,s}\}_{s\in S_t}$;
  Choose action $s_t^{\text{on}} \leftarrow \arg\max_{s\in S_{t,\Theta_t}} \{X_{t,s}\}$, breaking ties arbitrarily;
  Update $B_{t-1,i} \leftarrow B_{t,i} - a_{is_t^{\text{on}}}$ for each $i \in [M]$;
end
Update $\{S_{R,j}\}_{j\in[N]}$ and observe the last batch $\Theta_R, \Theta_{R-1}, \ldots, \Theta_1$; follow the offline optimal hindsight policy for the last batch.

We next present an upper bound on the expected regret of Algorithm 4. Recall that $\rho_1 = \exp(-2\lambda_{\min}^2)$, $\rho_2 = \lambda_{\min}\exp\left(-\frac{4\lambda_{\min}}{\kappa_{\max}^2(C_{\max}+1)^2}\right) + (1-\lambda_{\min})$ and $C_1 = (a_{\max}+1)C_{\max} + a_{\max}$. We further define

$$\rho_3 \triangleq \exp\left(-\frac{2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right), \quad \rho \triangleq \max\{\rho_1^2, \rho_3\},$$
$$c_1 = c_1(T,K) \triangleq \min\left\{\left\lceil\frac{T}{K+1}\right\rceil - 1,\ \left\lceil\frac{C_1}{\lambda_{\min}(K+1)}\right\rceil\right\},$$
$$c_2 = c_2(T,K) \triangleq 1 - \rho_3^{(K+1)\left(\left\lceil\frac{T}{K+1}\right\rceil-1\right)} = 1 - \rho_3^{T-R}.$$

THEOREM 4. *For any instance $\mathcal{I}$ and $K \geq 0$, the expected regret of Algorithm 4 has the following upper bound:*

$$a_{\max}\cdot M\cdot r_{\max}\cdot\left(c_1\cdot\sum_{m=0}^{K}\rho_1^{m-2\frac{C_1}{\lambda_{\min}}} + \frac{2c_2\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)}{1-\rho_3}\cdot\sum_{m=0}^{K}\rho_2^m\right)\cdot\rho^R.$$

Although c_1 and c_2 are functions of T, they are both bounded above by a constant that does not depend on T. As a consequence, Theorem 4 establishes a constant regret bound for the K-period batching model that is independent of T. In addition, when $K \geq T-1$, both c_1 and c_2 equal zero, implying that the regret upper bound is zero when there is only one batch. Moreover, the bound decays exponentially w.r.t. the length of the last batch R, where R can lie anywhere between 1 and $K+1$. This implies that batching is a weaker form of delay, which can also be illustrated through single-period regrets. Specifically, similar

to (17) for the multisecretary problem, we can derive the corresponding regret bound for periods $t = T, \ldots, R+1$:

$$r_N \cdot e^{-\lambda_{\min}} \cdot \left(\lambda_{\min} e^{-\lambda_{\min}} + (1-\lambda_{\min})\right)^{m(K,t)} \cdot \exp\left(-\frac{\lambda_{\min}^2}{2}\left(t - m(K,t) - 1\right)\right). \tag{29}$$

Note that $m(K,t)$ must be an integer within the set $\{0, 1, \ldots, K\}$. Thus, the single-period regret bound in (29) equals exactly $\mathrm{RB}(t,K)$ as given in (17) for periods where $m(K,t) = K$, and equals $\mathrm{RB}(t,0)$ for periods where $m(K,t) = 0$. The exponential decay in R can also be observed from (29) using the fact that $t - m(K,t) - 1 \geq R$ for all $t = T, \ldots, R+1$. Although our analysis focuses only on the upper bounds rather than the true regret values, we expect that the performance of K-period batching will be inferior to that of K-period delay, but superior to no delay. We confirm this expectation through the numerical experiments presented in Sections 7.2 and 7.3.

7. Numerical Experiments

In this section, we conduct three sets of numerical experiments. In Section 7.1, we evaluate the benefits of delay in the multisecretary problem by using synthetic data. We perform additional experiments in Section 7.2 to incorporate unknown arrival distribution and batching into the multisecretary problem. In Section 7.3, we evaluate the benefits of delay in an order-fulfillment problem by using a dataset of JD.com. The code is publicly available at `https://github.com/xieyaqi188/fulfillment_delay`.

7.1. Multisecretary Problem

In this section, we numerically evaluate the benefits of delay in the multisecretary problem by using synthetic data. We also illustrate the importance of utilizing all available information. Specifically, even when $K = 0$, our acceptance criterion (9), which leverages the available information from the current arrival, is superior to the standard acceptance criterion (10).

Recall that the multisecretary problem involves four parameters: the number of available positions B, the rewards $(r_j)_{j\in[N]}$, the arrival rates $\{\lambda_j\}_{j\in[N]}$, and the length of the time horizon T. In our experiments, we choose $N = 10$, $(r_j)_{j\in[N]} = (10, 20, 30, 40, 50, 60, 70, 80, 90, 100)$, $T = 100$, and test three different values for B in $\{25, 50, 75\}$. We also randomly generate 100 distributions, each defined by a vector $\{\lambda_j\}_{j\in[N]}$, which corresponds to a random point in the simplex where $\sum_{j\in[N]} \lambda_j = 1$. Using

different vectors for $\{\lambda_j\}_{j\in[N]}$ allows us to capture heterogeneity in the variance of distributions. We evaluate the benefits of delay based on different values of $K \in \{-1, 0, 1, \ldots, 10\}$, where $K = 0$ and $K = -1$ correspond to our online LP without delay (i.e., the one with the acceptance criterion (9)) and the classical LP (i.e., the one with the acceptance criterion (10)), respectively. For each of the 100 distributions, we generate 100 sample paths and record the regret ratios compared to the offline benchmark. We present the average ratios among the 10,000 samples for different values of K and B in Figure 1. Specifically, without any delay, $K = 0$ already reduces the average ratio by almost 50% compared with $K = -1$, confirming the intuition that our online LP better utilizes information to make improved decisions. Furthermore, $K = 1, 2$ further reduce the average ratio by another 50% compared with $K = 0$. Overall, the marginal benefit of delay, defined as the marginal decrease in the average regret ratio, is decreasing in K (i.e., the regret ratio is "convex" in K in Figure 1). Thus, the results demonstrate that a little delay is all we need.

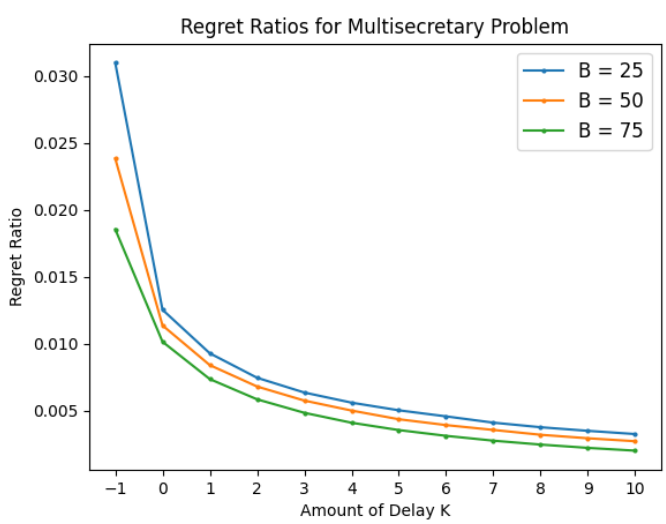

Figure 1 **Regret ratios for Algorithm 1 on multisecretary problem.**

7.2. Unknown Arrival Distribution and Batching in the Multisecretary Problem

In this section, we conduct additional experiments in the multisecretary problem based on the extensions in Section 6. Specifically, we evaluate the benefits of delay under unknown arrival distributions and compare these benefits with those of batching.

For the setting with unknown arrival distributions, we use the same parameter setup as in Section 7.1 but do not provide the algorithm with the arrival rates. We vary the length of the delay $K \in \{0, 1, \ldots, 20\}$ and record the average performance ratio across 10,000 samples for each K and starting inventory B. Recall that the marginal benefit of delay is defined as the marginal decrease between two consecutive regret ratios; more specifically, the marginal benefit of a K-period delay is defined as the difference in regret ratios between a $(K-1)$-period delay and a K-period delay. We summarize the results in Figure 2. The marginal benefits are consistently higher in the setting with unknown distribution than in the setting with known distribution, demonstrating that delay provides the additional benefit of helping the algorithm learn when the arrival distribution is initially unknown.

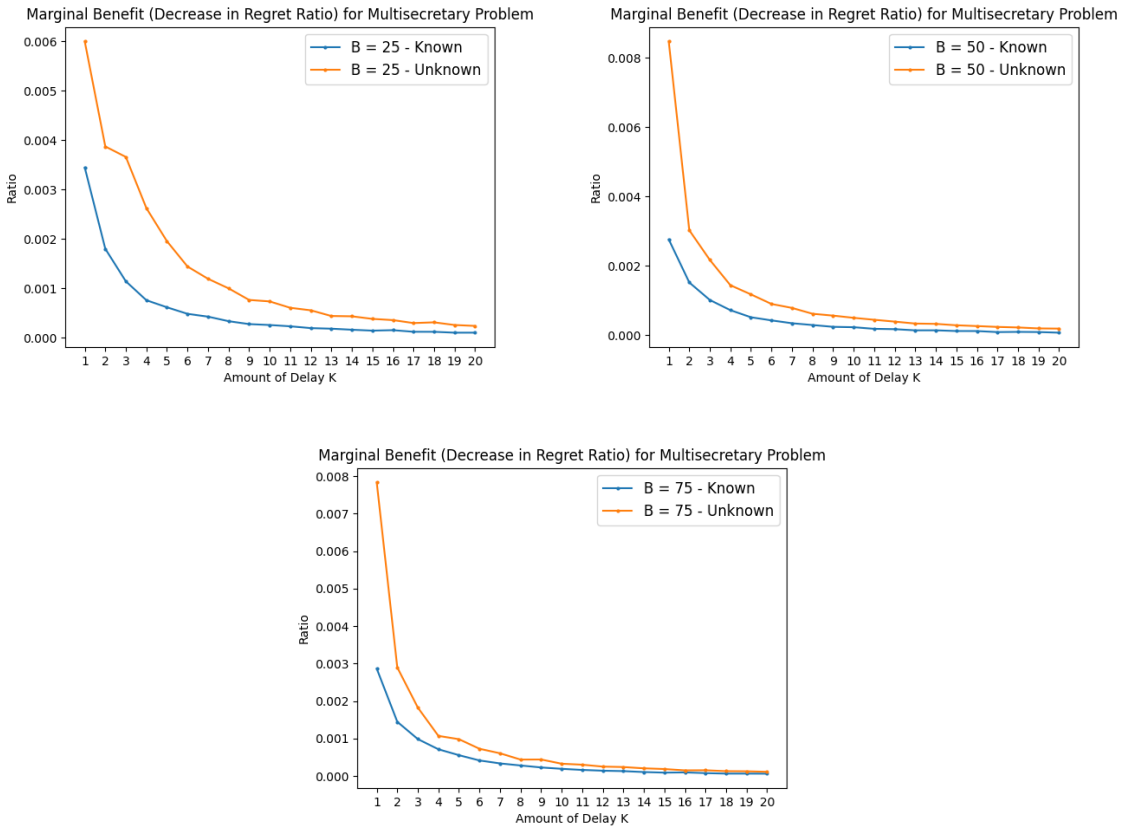

Figure 2 **Comparing the marginal benefit of delay for Algorithm 2 (operating under known distributions) vs. Algorithm 3 (operating under unknown distributions).**

For batching, we also use the same parameter setup as in Section 7.1. Recall that "K-period batching" refers to batches of size $K+1$ (with 0-period batching representing

the original setting with no delay or batching). We present the average regret ratios for batching, again using 10,000 samples for each K and starting inventory B, in Figure 3. The results suggest that the marginal benefit of batching also decreases with K, consistent with the results for delay. We also compare the benefits of batching with those of delay. Figure 3 demonstrates that delay consistently offers higher benefits, which aligns with the observation that batching is a weaker form of delay. Finally, it is worth noting that the regret ratio for batching is not necessarily monotonic for large values of K. This is because the impact of batching is sensitive to the length of the last batch.

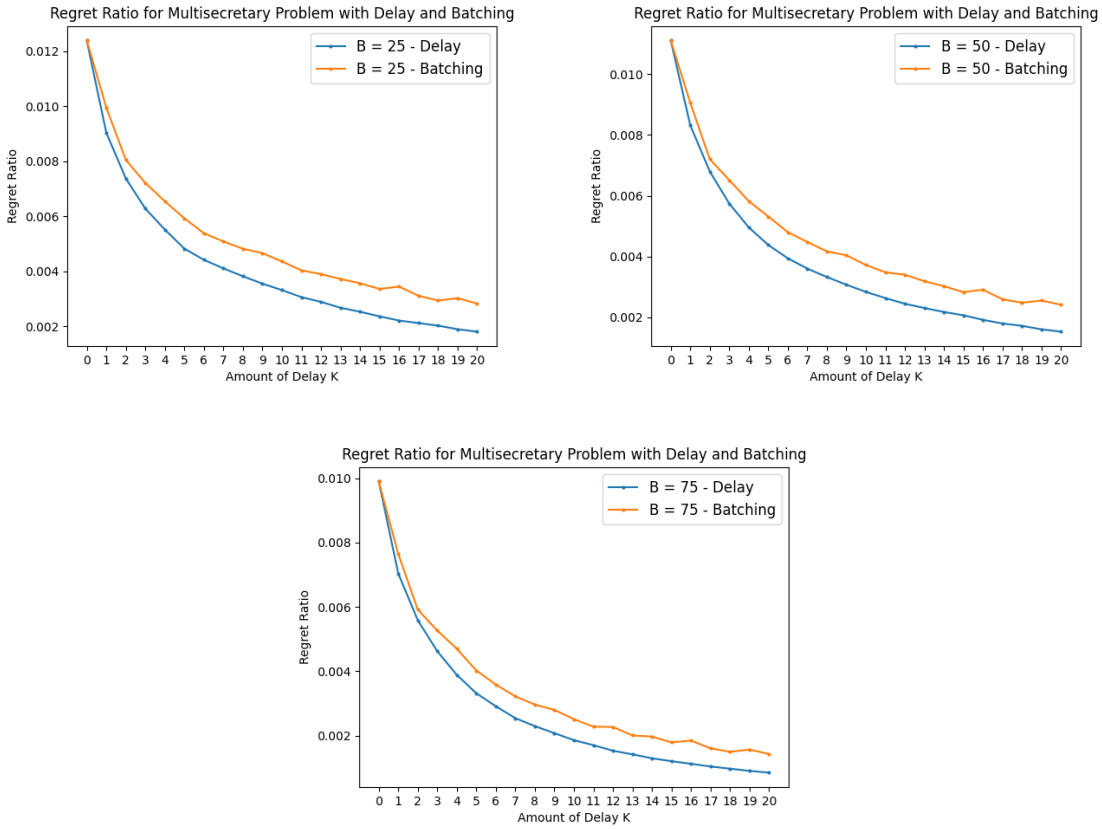

Figure 3 **Regret ratios for Algorithm 2 (delay) vs. Algorithm 4 (batch) on multisecretary problem.**

7.3. Order-Fulfillment Problem

In this section, we use a dataset from JD.com (Shen et al. 2024) to evaluate the benefits of delay in an order-fulfillment problem. Let us begin with a brief description of JD.com's two-layer fulfillment network, which includes front distribution centers (FDCs) and regional

distribution centers (RDCs). FDCs are warehouses located closer to end customers but have limited capacities, whereas RDCs are larger warehouses that replenish the lower-layer FDCs and serve as a "back-up" option when FDCs run out of inventory. A unique challenge in e-commerce fulfillment arises from multi-item orders (i.e., orders consisting of more than one item), which can lead to split orders and/or shipping from distant warehouses. In fact, one of JD.com's key performance metrics for fulfillment is the percentage of orders that are entirely fulfilled by local FDCs. We refer interested readers to Shen et al. (2024) and Qin et al. (2022) for more details about JD.com's fulfillment operations.

Following Zhao et al. (2020), we consider a region consisting of one FDC with limited inventory and one RDC with infinite inventory. There are M types of stock-keeping units (SKUs) indexed by i, and their initial inventories at the FDC are $B = (B_1, \ldots, B_M)$. There are N types of orders indexed by j. For each arriving order of type j, the decision is either to accept the order at the FDC (if possible) which would require a_{ij} units of SKU i for all $i \in [M]$, or to reject the order (i.e., fulfill it from the RDC). It is unnecessary to consider the option of splitting an order between the FDC and RDC due to the infinite inventory at the RDC. The objective is to maximize the number of orders served by the FDC. We illustrate the benefits of delay in this setting with the following toy example. Suppose there are four SKUs: {A, B, C, D}, and each has one unit of inventory at the FDC. Three orders arrive in sequence: the first order contains one SKU B and one SKU C; the second order contains one SKU A and one SKU B; and the third order contains one SKU C and one SKU D. A greedy-type online policy without delay fulfills the first order from the FDC and the remaining two from the RDC, whereas a delayed policy fulfills the last two orders from the FDC and the first one from the RDC. As such, the delayed policy can fulfill one more order from the FDC.

We next describe the dataset, which contains orders from the month of March 2018. For each order, the transaction-level data includes the order time, SKU IDs and corresponding quantities, the origin center, and the destination center. We ignore third-party-owned SKUs and focus only on SKUs managed by JD.com, in which case approximately 20% of orders are multi-item or multi-unit. We construct order types based on the SKU IDs and associated quantities. For each transaction, the origin center is the warehouse from which the order is shipped, and the destination center is the designated warehouse responsible for fulfilling orders from the customer's district. Ideally the origin and destination centers

are the same. If not, it usually indicates that shipping is not from the nearest warehouse. Finally, the data provides the information on relationships between FDCs and RDCs (i.e., which RDC is supposed to fulfill an order when the designated FDC is out of stock).

We treat the order sequence on a given date as a sample path, with 31 sample paths for each district. For each district, we include order records where the associated FDC and its RDC serve as the origin centers, and the FDC as the destination center, as part of the order sequence. For each FDC-SKU combination, we set the initial inventory by using the average daily sales with the FDC as the origin center. To generate forecasts for our algorithm, we divide the hours in a day into 48 half-hour time slots. For each time slot, we set the arrival rates of order types by using the average sales with the FDC or its RDC as the origin center. We also consider time-varying arrival rates between weekdays and weekends. We evaluate the benefits of delay and batching for durations of 30 minutes, 60 minutes, 90 minutes, 120 minutes, 180 minutes, and 240 minutes. For a K-minute delay, we solve an online LP for the order arriving at time t based on the realized demand from t to $t+K$ and the expected demand afterward. Here, all orders are delayed by K minutes (not by K orders), which our model captures by interpreting t as a short time duration during which at most one arrival occurs and by introducing a "null" type to represent no arrival. For K-minute batching, we divide the day into $1440/K$ batches, and solve the corresponding online LP for the order arriving at time t based on the realized demand from t until the end of the batch containing that order, as well as the expected demand in the remaining batches. The offline benchmark in the simulation is solved by using an integer program. For each district, we record the average ratios of the regrets from the two online algorithms to the reward of the offline benchmark across the 31 sample paths observed in practice. We then present the average ratios for the 20 districts with the highest sales in Figure 4. Note that in practical applications, the regret curves shown in Figure 4 enable decision-makers to establish quantitative relationships between delay length and the reduction in regret. This allows them to determine the optimal delay length by combining these curves with the costs incurred by the delay. In particular, with a 60-minute delay, the ratio decreases by almost 1.5%, which further confirms that a little delay brings the most benefits. Here, we do not expect a more substantial percentage reduction in such an RDC-FDC fulfillment network, because an online policy without delay is already expected to perform well (e.g., Zhao et al. 2020). However, this 1.5% improvement still represents a substantial operational

gain, especially given the large volume and the low cost of implementing a 60-minute delay. In addition, as previously discussed, Figure 4 demonstrates that K-period batching performs worse than K-period delay but better than the no-delay setting. We also observe that the ratio is convex w.r.t. the length of delay, which is consistent with the intuition that the marginal benefit decreases as the length of delay increases.

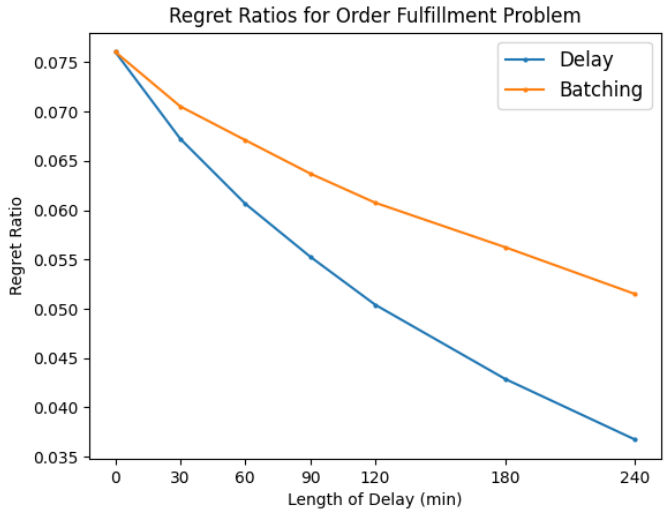

Figure 4 **Regret ratios for Algorithm 2 (delay) vs. Algorithm 4 (batching) in the order fulfillment problem.**

8. Conclusion

In this paper, we theoretically characterized the benefits of delaying real-time decisions for a broad family of online decision-making problems. We proved that the gap between the proposed delayed Bayesian prophet algorithm and the offline optimal hindsight policy decays exponentially fast in the length of delay. We also conducted extensive numerical experiments to assess the benefits of delay by using both synthetic and real data. Both our theoretical and empirical results conveyed an important insight: a little delay is all we need. Finally, we extended our analysis and results to the settings with non-IID and unknown arrival distributions and to those where decisions are made in batches.

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Supplementary Material

EC.1. Proof of Lemma 1

Proof of Lemma 1. Recall that $V^{\text{on}}(\mathcal{I}) = \sum_{t=1}^{T} r_{s_t^{\text{on}}}$. We decompose $\text{REG}_0 = \mathbb{E}_\Theta\left[V^{\text{off}}(\mathcal{I},\Theta,B)\right] - V^{\text{on}}(\mathcal{I})$ in (3) into the regret from time period T to period $K+2$ and the regret during the last $K+1$ time periods:

$$\begin{aligned}
&\mathbb{E}_\Theta\left[V^{\text{off}}(\mathcal{I},\Theta,B_T) - V^{\text{off}}(\mathcal{I},(\Theta_{K+1},\ldots,\Theta_1),B_{K+1}) + V^{\text{off}}(\mathcal{I},(\Theta_{K+1},\ldots,\Theta_1),B_{K+1}) - \sum_{t=1}^{T} r_{s_t^{\text{on}}}\right] \\
&= \sum_{t=K+2}^{T} \mathbb{E}_{(\Theta_t,\ldots,\Theta_1),B_t}\left[V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t) - V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1}) - r_{s_t^{\text{on}}}\right] \\
&\quad + \mathbb{E}_{(\Theta_{K+1},\ldots,\Theta_1),B_{K+1}}\left[V^{\text{off}}(\mathcal{I},(\Theta_{K+1},\ldots,\Theta_1),B_{K+1}) - \sum_{t=1}^{K+1} r_{s_t^{\text{on}}}\right] \\
&\le a_{\max}\cdot M\cdot r_{\max}\cdot \sum_{t=K+2}^{T} \mathbb{P}\left(V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t) - V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1}) > r_{s_t^{\text{on}}}\right) + \sum_{s\in S} r_s.
\end{aligned}$$

In particular, the first part of the last inequality comes from the explanation right below the statement of Lemma 1, and the second part follows from the fact that

$$V^{\text{off}}(\mathcal{I},(\Theta_{K+1},\ldots,\Theta_1),B_{K+1}) - \sum_{t=1}^{K+1} r_{s_t^{\text{on}}} \le \sum_{s\in S} r_s. \tag{EC.1}$$

To see (EC.1), note that with a K-period delay, the online policy has full knowledge of $(\Theta_{K+1},\ldots,\Theta_1)$ so that the gap between $V^{\text{off}}(\mathcal{I},(\Theta_{K+1},\ldots,\Theta_1),B_{K+1})$ and $\sum_{t=1}^{K+1} r_{s_t^{\text{on}}}$ is due to integrality. This gap has an upper bound $\sum_{s\in S} r_s$ by rounding each variable of the optimal solution to the LP, which is independent of T and K. The proof of Lemma 1 is completed. □

EC.2. Proofs of Statements in the Multisecretary Problem

Derivation of (8). We first characterize the optimal solutions to the offline and online LP Problems (6) and (7). Because both B_t and $D_j[t,1]$ are integers, Problem (6) has an optimal integer solution $\left\{X^*_{t,j},\ X^*_{t,\emptyset j}\right\}_{j\in[N]}$ for each t:

$$X^*_{t,j} = \min\left\{\left(B_t - \sum_{\ell=j+1}^{N} D_\ell[t,1]\right)^+,\ D_j[t,1]\right\} \quad \text{and} \quad X^*_{t,\emptyset j} = D_j[t,1] - X^*_{t,j}. \tag{EC.2}$$

Similarly, Problem (7) has an optimal solution $\{X_{t,j},\ X_{t,\emptyset j}\}_{j\in[N]}$:

$$X_{t,j} = \min\left\{\left(B_t - \sum_{\ell=j+1}^{N}\left(D_\ell[t,t-K]+\lambda_\ell[t-K-1,1]\right)\right)^+,\ D_j[t,t-K]+\lambda_j[t-K-1,1]\right\}, \tag{EC.3}$$

$$X_{t,\emptyset j} = D_j[t,t-K]+\lambda_j[t-K-1,1]-X_{t,j}.$$

Recall that for each $t=T,T-1,\ldots,K+2$, the algorithm accepts the candidate with type Θ_t at time t iff $X_{t,\Theta_t}\geq X_{t,\emptyset\Theta_t}$, which is equivalent to that at least half of type Θ_t candidates are accepted in the LP solution:

$$X_{t,\Theta_t} \geq \frac{1}{2}\left(D_{\Theta_t}[t,t-K]+\lambda_{\Theta_t}[t-K-1,1]\right). \tag{EC.4}$$

Combining with (EC.3), (EC.4) is equivalent to

$$\begin{aligned} B_t &\geq \sum_{\ell=\Theta_t+1}^{N}\left(D_\ell[t,t-K]+\lambda_\ell[t-K-1,1]\right)+\frac{1}{2}\left(D_{\Theta_t}[t,t-K]+\lambda_{\Theta_t}[t-K-1,1]\right)\\ &= \left(1-F_{\Theta_t}+\frac{\lambda_{\Theta_t}}{2}\right)(t-K-1)+\left(\sum_{\ell=\Theta_t+1}^{N}D_\ell[t,t-K]+\frac{1}{2}\cdot D_{\Theta_t}[t,t-K]\right). \end{aligned}$$

Similarly, the algorithm rejects the candidate with type Θ_t at time t iff

$$X_{t,\Theta_t} < \frac{1}{2}\left(D_{\Theta_t}[t,t-K]+\lambda_{\Theta_t}[t-K-1,1]\right). \tag{EC.5}$$

Combining with (EC.3), (EC.5) is equivalent to

$$B_t < \left(1-F_{\Theta_t}+\frac{\lambda_{\Theta_t}}{2}\right)(t-K-1)+\left(\sum_{\ell=\Theta_t+1}^{N}D_\ell[t,t-K]+\frac{1}{2}\cdot D_{\Theta_t}[t,t-K]\right).$$

The derivation of (8) is completed. □

Proof of Theorem 1. Combining (5) with the fact that $a_{\max}=1$, $M=1$, $r_{\max}=r_N$, and when the decision for Θ_t incurs regret is exactly when incorrect acceptance or rejection occurs, the regret REG of Algorithm 1 can be upper bounded by

$$r_N\cdot\sum_{t=K+2}^{T}\mathbb{P}\left(V^{\text{off}}\left(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t\right)-V^{\text{off}}\left(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1}\right)>r_{s_t^{\text{on}}}\right). \tag{EC.6}$$

Note that the probability in (EC.6) is w.r.t. $(\Theta_{t-1},\ldots,\Theta_1)$ and B_t. It is straightforward to verify that incorrect acceptance for Θ_t occurs iff $X^*_{t,\Theta_t}=0$ and (8) holds, and incorrect rejection for Θ_t occurs iff $X^*_{t,\emptyset\Theta_t}=0$ and the reverse of (8) holds. From (EC.2),

$$\begin{aligned} X^*_{t,\Theta_t}=0 &\iff \sum_{\ell=\Theta_t+1}^{N} D_\ell[t,1]\geq B_t,\\ X^*_{t,\emptyset\Theta_t}=0 &\iff \sum_{\ell=\Theta_t}^{N} D_\ell[t,1]\leq B_t. \end{aligned} \tag{EC.7}$$

Combining (EC.7) and (8), it follows that incorrect acceptance implies

$$\sum_{\ell=\Theta_t+1}^{N} D_\ell[t-K-1,1]-(1-F_{\Theta_t})(t-K-1)\geq \frac{\lambda_{\Theta_t}}{2}\cdot(t-K-1)+\frac{1}{2}\cdot D_{\Theta_t}[t,t-K],$$

and incorrect rejection implies

$$(1-F_{\Theta_t}+\lambda_{\Theta_t})\,(t-K-1)-\sum_{\ell=\Theta_t}^{N} D_\ell[t-K-1,1]\geq \frac{\lambda_{\Theta_t}}{2}\cdot(t-K-1)+\frac{1}{2}\cdot D_{\Theta_t}[t,t-K].$$

Hence, conditional on $(\Theta_t,\ldots,\Theta_{t-K})$, the probability of incorrect acceptance for Θ_t is at most

$$\mathbb{P}\left(\sum_{\ell=\Theta_t+1}^{N} D_\ell[t-K-1,1]-(1-F_{\Theta_t})(t-K-1)\geq \frac{\lambda_{\Theta_t}}{2}\cdot(t-K-1)+\frac{1}{2}\cdot D_{\Theta_t}[t,t-K]\right), \tag{EC.8}$$

and the probability of incorrect rejection for Θ_t is at most

$$\mathbb{P}\left((1-F_{\Theta_t}+\lambda_{\Theta_t})\,(t-K-1)-\sum_{\ell=\Theta_t}^{N} D_\ell[t-K-1,1]\geq \frac{\lambda_{\Theta_t}}{2}\cdot(t-K-1)+\frac{1}{2}\cdot D_{\Theta_t}[t,t-K]\right). \tag{EC.9}$$

Furthermore, from Hoeffding's inequality, both (EC.8) and (EC.9) can be upper bounded by

$$\exp\left(-\frac{t-K-1}{2}\cdot\left(\lambda_{\Theta_t}+\frac{D_{\Theta_t}[t,t-K]}{t-K-1}\right)^2\right). \tag{EC.10}$$

Note that B_t is the only state variable in the multisecretary problem and the upper bound (EC.10) holds for any value of B_t. Thus, conditional on Θ_t, it follows that

$$\begin{aligned} &\mathbb{P}\left(V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t)-V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1})>r_{s_t^{\text{on}}}\right)\\ &\leq \mathbb{E}\left[\exp\left(-\frac{t-K-1}{2}\cdot\left(\lambda_{\Theta_t}+\frac{D_{\Theta_t}[t,t-K]}{t-K-1}\right)^2\right)\middle|\Theta_t\right], \end{aligned} \tag{EC.11}$$

where the probability and expectation above are w.r.t. $(\Theta_{t-1},\ldots,\Theta_{t-K})$. Furthermore,

$$
\begin{aligned}
&\mathbb{E}\left[\exp\left(-\frac{t-K-1}{2}\cdot\left(\lambda_{\Theta_t}+\frac{D_{\Theta_t}[t,t-K]}{t-K-1}\right)^2\right)\middle|\Theta_t\right]\\
=\;&\exp\left(-\frac{\lambda_{\Theta_t}^2}{2}\cdot(t-K-1)\right)\cdot\mathbb{E}\left[\exp\left(-\lambda_{\Theta_t}\cdot D_{\Theta_t}[t,t-K]\right)\cdot\exp\left(-\frac{\left(D_{\Theta_t}[t,t-K]\right)^2}{2\left(t-K-1\right)}\right)\middle|\Theta_t\right]\\
\le\;&\exp\left(-\frac{\lambda_{\Theta_t}^2}{2}\cdot(t-K-1)\right)\cdot\mathbb{E}\left[\exp\left(-\lambda_{\Theta_t}\cdot D_{\Theta_t}[t,t-K]\right)\middle|\Theta_t\right]\\
=\;&\exp\left(-\frac{\lambda_{\Theta_t}^2}{2}\cdot(t-K-1)\right)\cdot e^{-\lambda_{\Theta_t}}\cdot\mathbb{E}\left[\exp\left(-\lambda_{\Theta_t}\cdot D_{\Theta_t}[t-1,t-K]\right)\middle|\Theta_t\right] \qquad \text{(EC.12)}\\
=\;&\exp\left(-\frac{\lambda_{\Theta_t}^2}{2}\cdot(t-K-1)\right)\cdot e^{-\lambda_{\Theta_t}}\cdot\left(\lambda_{\Theta_t}e^{-\lambda_{\Theta_t}}+(1-\lambda_{\Theta_t})\right)^K, \qquad \text{(EC.13)}
\end{aligned}
$$

where the inequality comes from the fact that the term $\exp\left(-\frac{\left(D_{\Theta_t}[t,t-K]\right)^2}{2(t-K-1)}\right)$ is no greater than one and (EC.12) is from the fact that $D_{\Theta_t}[t,t]=1$ conditional on Θ_t.

Recall that $\lambda_{\min}=\min_{j\in[N]}\lambda_j$ and $\rho_0=\lambda_{\min}e^{-\lambda_{\min}}+(1-\lambda_{\min})<1$. Note that the function $g(x)\triangleq xe^{-x}+(1-x)$ is non-increasing in x on $[0,1]$. Hence, combining (EC.6), (EC.11), and (EC.13), the expected regret is no greater than

$$
\begin{aligned}
&r_N\cdot\sum_{t=K+2}^{T}\sum_{j\in[N]}\lambda_j\left(e^{-\lambda_j}\cdot\exp\left(-\frac{\lambda_j^2}{2}\cdot(t-K-1)\right)\cdot\left(\lambda_je^{-\lambda_j}+(1-\lambda_j)\right)^K\right)\\
\le\;&r_N\cdot\sum_{t=K+2}^{T}\sum_{j\in[N]}\lambda_j\left(e^{-\lambda_{\min}}\cdot\exp\left(-\frac{\lambda_{\min}^2}{2}\cdot(t-K-1)\right)\cdot\left(\lambda_{\min}e^{-\lambda_{\min}}+(1-\lambda_{\min})\right)^K\right)\\
=\;&r_N\cdot e^{-\lambda_{\min}}\cdot\rho_0^K\cdot\sum_{t=K+2}^{T}\exp\left(-\frac{\lambda_{\min}^2}{2}\cdot(t-K-1)\right)\\
\le\;&r_N\cdot e^{-\lambda_{\min}}\cdot\rho_0^K\cdot\sum_{t=1}^{\infty}\exp\left(-\frac{\lambda_{\min}^2t}{2}\right)\\
\le\;&r_N\cdot e^{-\lambda_{\min}}\cdot\rho_0^K\cdot\frac{\exp(-\lambda_{\min}^2/2)}{1-\exp(-\lambda_{\min}^2/2)}\\
=\;&r_N\cdot e^{-\lambda_{\min}}\cdot\rho_0^K\cdot\frac{1}{\exp(\lambda_{\min}^2/2)-1}\\
\le\;&\rho_0^K\cdot\frac{2\cdot r_N\cdot e^{-\lambda_{\min}}}{\lambda_{\min}^2},
\end{aligned}
$$

where the last inequality comes from the fact that $e^x\ge 1+x$ for $x\ge 0$. The proof of Theorem 1 is completed. □

EC.3. Proofs of Statements in the General Online Resource Allocation Problem

Proof of Theorem 2. We first prove that $X^*_{t,s_t^{\text{on}}} < a_{\max}+1$ is a necessary condition for

$$V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t) - V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1}) > r_{s_t^{\text{on}}}.$$

Indeed, if $X^*_{t,s_t^{\text{on}}} \geq a_{\max}+1$, then $B_{t,i}$ for each resource $i\in[M]$ satisfying $a_{is_t^{\text{on}}}>0$ is at least $(a_{\max}+1)a_{is_t^{\text{on}}}$. This implies that the remaining resource after action s_t^{on} is at least $a_{\max}\cdot a_{is_t^{\text{on}}} \geq a_{\max}$, because $a_{\min}$ is normalized to be 1. Thus, action s_t^{on} does not result in an update of feasible action sets. That is, $S^t = S^{t-1}$, and $V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t)$ and $V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1})$ have the same coefficient matrix. From this, we prove that $V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t) = V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1}) + r_{s_t^{\text{on}}}$. More specifically, for any given sample path Θ, we define $\omega^{\text{on}}_{t,s}$ as the indicator of whether action $s\in S_j$ is chosen for arrival Θ_t with customer type j, namely, $\omega^{on}_{t,s} = \mathbb{I}\{\Theta_t = j, s_t^{\text{on}} = s\}$. Let us consider the following variant of the offline Problem (19):

$$\begin{aligned}
\max \quad & \sum_{s\in S_t} r_s x_s && && \text{(EC.14)}\\
\text{s.t.} \quad & \sum_{s\in S_t} a_{is}x_s \leq B_{t,i}, && \forall i\in[M],\\
& \sum_{s\in S_{t,j}} x_s = D_j[t,1], && \forall j\in[N],\\
& x_s \geq \omega_s, && \forall s\in S_t.
\end{aligned}$$

Let $\hat{V}^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t,\{\omega_s\})$ denote the optimal value of Problem (EC.14) with right-hand side vector B_t, $\{D_j[t,1]\}$, and $\{\omega_s\}$. It is easy to see that the optimal value of the offline LP Problem (19) is $\hat{V}^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t,0) = V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t)$. After action s_t^{on} is chosen for Θ_t, the optimal value that can be obtained afterwards reduces to $\hat{V}^{\text{off}}\left(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t,\{\omega^{\text{on}}_{t,s}\}\right)$, Note that $\hat{V}^{\text{off}}\left(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t,\{\omega^{\text{on}}_{t,s}\}\right) = V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1}) + r_{s_t^{\text{on}}}$, because $S^t = S^{t-1}$. Thus, for any given t, the regret of the decision s_t^{on} for Θ_t is

$$\begin{aligned}
\text{Diff} \triangleq\ & \hat{V}^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t,0) - \hat{V}^{\text{off}}\left(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t,\{\omega^{\text{on}}_{t,s}\}\right)\\
=\ & V^{\text{off}}(\mathcal{I},(\Theta_t,\ldots,\Theta_1),B_t) - \left(V^{\text{off}}(\mathcal{I},(\Theta_{t-1},\ldots,\Theta_1),B_{t-1}) + r_{s_t^{\text{on}}}\right).
\end{aligned}$$

Furthermore, for an optimal solution $\{X^*_{t,s}\}_{s\in S_t}$ of Problem (19), because $X^*_{t,s^{\text{on}}_t} \geq a_{\max}+1 \geq 1$, we have that $\{X^*_{t,s}\}_{s\in S_t}$ is a feasible solution of Problem (EC.14) with $\omega_s = \omega^{\text{on}}_{t,s}$, implying $\text{Diff} = 0$. Therefore, if $\text{Diff} > 0$, then we must have $X^*_{t,s^{\text{on}}_t} < a_{\max}+1$.

Plugging this fact into (5), we have

$$\begin{aligned}\text{REG} \leq\ & a_{\max}\cdot M\cdot r_{\max}\cdot \sum_{t=K+2}^{T} \mathbb{P}\left(X^*_{t,s^{\text{on}}_t} < a_{\max}+1\right)\\ =\ & a_{\max}\cdot M\cdot r_{\max}\cdot \sum_{t=K+2}^{T}\sum_{j\in[N]} \lambda_j\cdot \mathbb{P}\left(X^*_{t,s^{\text{on}}_t} < a_{\max}+1 | \Theta_t = j\right). \end{aligned} \tag{EC.15}$$

Recall that $C_1 = (a_{\max}+1)C_{\max} + a_{\max}$. We next bound $\mathbb{P}\left(X^*_{t,s^{\text{on}}_t} < a_{\max}+1|\Theta_t = j\right)$:

$$\begin{aligned}&\mathbb{P}\left(X^*_{t,s^{\text{on}}_t} < a_{\max}+1|\Theta_t = j\right)\\ \leq\ & \mathbb{P}\left(X^*_{t,s^{\text{on}}_t} < a_{\max}+1+X_{t,s^{\text{on}}_t} - \frac{D_j[t,t-K]+\lambda_j(t-K-1)}{C_{\max}+1}\middle|\Theta_t = j\right)\\ \leq\ & \mathbb{P}\left(\max_{s\in S_t}\left|X^*_{t,s}-X_{t,s}\right| > \frac{D_j[t,t-K]+\lambda_j(t-K-1)}{C_{\max}+1} - (a_{\max}+1)\middle|\Theta_t = j\right)\\ \leq\ & \mathbb{P}\left(\max_{j\in[N]}\left|D_j[t-K-1,1]-\lambda_j(t-K-1)\right| > \frac{D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1}{\kappa_t(C_{\max}+1)}\middle|\Theta_t = j\right)\\ \leq\ & \mathbb{P}\left(\left|D_{j^*_t}[t-K-1,1]-\lambda_{j^*_t}(t-K-1)\right| > \frac{D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1}{\kappa_{\max}(C_{\max}+1)}\middle|\Theta_t = j\right)\\ \triangleq\ & \mathbb{P}\left(E^j_t\right). \end{aligned} \tag{EC.16}$$

Here, $\kappa_t = \kappa(J_t)$, $j^*_t \in \arg\max_{j\in[N]}\{|D_j[t-K-1,1]-\lambda_j(t-K-1)|\}$, and E^j_t represents the following event conditional on $\Theta_t = j$:

$$\left|D_{j^*_t}[t-K-1,1]-\lambda_{j^*_t}(t-K-1)\right| > \frac{D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1}{\kappa_{\max}(C_{\max}+1)}.$$

In addition, the first inequality comes from (21), the third inequality comes from (22), and the fourth inequality is due to the monotonicity of κ_t in t.

Note that (EC.16) is independent of B_t. Conditional on $(\Theta_{t-1},\ldots,\Theta_{t-K})$, we further decompose $\mathbb{P}\left(E^j_t|\Theta_{t-1},\ldots,\Theta_{t-K}\right)$ into two parts:

$$\mathbb{P}\left(E^j_t|\Theta_{t-1},\ldots,\Theta_{t-K}\right) = A^{j,1}_t + A^{j,2}_t, \tag{EC.17}$$

where

$$\begin{aligned} A_t^{j,1} \triangleq\ & \mathbb{I}\left(D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1<0\right) \\ & \cdot \mathbb{P}\left(E_t^j \middle| D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1<0\right), \\ A_t^{j,2} \triangleq\ & \mathbb{I}\left(D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1\geq 0\right) \\ & \cdot \mathbb{P}\left(E_t^j \middle| D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1\geq 0\right). \end{aligned}$$

It is easy to see that $\mathbb{P}\left(E_t^j \middle| D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1<0\right)=1$. It implies that

$$\begin{aligned} & \sum_{t=K+2}^{T} \mathbb{E}\left[A_t^{j,1}\right] \\ = & \sum_{t=K+2}^{T} \mathbb{E}\Bigg[\mathbb{I}\left(D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1<0\right) \\ & \qquad \cdot \mathbb{P}\left(E_t^j \middle| D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1<0\right)\Bigg] \\ = & \sum_{t=K+2}^{T} \mathbb{P}\left(D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1<0\right)\cdot 1, \\ = & \sum_{m=1}^{T-K-1} \mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)<C_1-\lambda_j m\right), \end{aligned}$$

where the expectation above is w.r.t. $(\Theta_{t-1},\ldots,\Theta_{t-K})$, and the last equality holds because $D_j[t-1,t-K]$ follows a binomial distribution with parameters K and λ_j (denoted by $\mathrm{Bin}(K,\lambda_j)$). It is straightforward to see that

$$\begin{aligned} \sum_{m=1}^{T-K-1} \mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)<C_1-\lambda_j m\right) \leq & \left\lceil \frac{C_1}{\lambda_j} \right\rceil \cdot \mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)\leq C_1\right) \\ \leq & \left\lceil \frac{C_1}{\lambda_{\min}} \right\rceil \cdot \mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)\leq C_1\right). \end{aligned} \tag{EC.18}$$

For $K>\frac{C_1}{\lambda_{\min}}$, Hoeffding's inequality can yield the following upper bound on $\mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)\leq C_1\right)$:

$$\begin{aligned} \mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)\leq C_1\right) \leq & \exp\left(-2K\left(\lambda_j-\frac{C_1}{K}\right)^2\right) \\ \leq & \exp\left(-2K\left(\lambda_{\min}-\frac{C_1}{K}\right)^2\right) \\ = & \exp\left(-2K\lambda_{\min}^2+4\lambda_{\min}C_1-2\frac{C_1^2}{K}\right) \end{aligned}$$

$$\leq \exp\left(-2\lambda_{\min}^2\left(K-2\frac{C_1}{\lambda_{\min}}\right)\right).$$

In addition, $\mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)\leq C_1\right)$ has a trivial upper bound 1 for $0\leq K\leq \frac{C_1}{\lambda_{\min}}$. Thus, for all $K\geq 0$, we have

$$\mathbb{P}\left(\mathrm{Bin}(K,\lambda_j)\leq C_1\right)\leq \left(\exp\left(-2\lambda_{\min}^2\right)\right)^{K-2\frac{C_1}{\lambda_{\min}}}=\rho_1^{K-2\frac{C_1}{\lambda_{\min}}}, \tag{EC.19}$$

where $\rho_1=\exp(-2\lambda_{\min}^2)$. It implies that

$$\sum_{t=K+2}^{T}\mathbb{E}\left[A_t^{j,1}\right]\leq\left\lceil\frac{C_1}{\lambda_{\min}}\right\rceil\cdot\rho_1^{K-2\frac{C_1}{\lambda_{\min}}}. \tag{EC.20}$$

Next, we bound $\sum_{t=K+2}^{T}\mathbb{E}\left[A_t^{j,2}\right]$ from above. From Hoeffding's inequality, for all $t\geq K+2$, we have

$$\begin{aligned}&\mathbb{P}\left(E_t^j\middle|D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1\geq 0\right)\\ &\leq 2\exp\left(-2\frac{\left(D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1\right)^2}{(t-K-1)\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right),\end{aligned}$$

which follows that

$$\begin{aligned}&\sum_{t=K+2}^{T}\mathbb{E}\left[A_t^{j,2}\right]\\ &\leq\sum_{t=K+2}^{T}\mathbb{E}\Bigg[\mathbb{I}\left(D_j[t-1,t-K]+\lambda_j(t-K-1)\geq C_1\right)\\ &\qquad\cdot 2\exp\left(-2\frac{\left(D_j[t-1,t-K]+\lambda_j(t-K-1)-C_1\right)^2}{(t-K-1)\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\Bigg]\\ &\leq 2\sum_{t=K+2}^{T}\mathbb{E}\left[\exp\left(-2\frac{\left(\lambda_j(t-K-1)+\left(D_j[t-1,t-K]-C_1\right)\right)^2}{(t-K-1)\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\right].\end{aligned}$$

In addition,

$$\begin{aligned}&\mathbb{E}\left[\exp\left(-2\frac{\left(\lambda_j(t-K-1)+\left(D_j[t-1,t-K]-C_1\right)\right)^2}{(t-K-1)\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\right]\\ &=\exp\left(-\frac{2\lambda_j^2(t-K-1)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\mathbb{E}\Bigg[\exp\left(-\frac{4\lambda_j\left(D_j[t-1,t-K]-C_1\right)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\\ &\qquad\cdot\exp\left(-\frac{2\left(D_j[t-1,t-K]-C_1\right)^2}{(t-K-1)\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\Bigg]\end{aligned}$$

$$
\begin{aligned}
&\leq \exp\left(-\frac{2\lambda_j^2(t-K-1)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\mathbb{E}\left[\exp\left(-\frac{4\lambda_j\left(D_j[t-1,t-K]-C_1\right)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\right]\\
&= \exp\left(-\frac{2\lambda_j^2(t-K-1)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\exp\left(\frac{4\lambda_j\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\left(\lambda_j\exp\left(-\frac{4\lambda_j}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)+(1-\lambda_j)\right)^K\\
&\leq \exp\left(-\frac{2\lambda_{\min}^2(t-K-1)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\rho_2^K, \qquad \text{(EC.21)}
\end{aligned}
$$

where $\rho_2=\lambda_{\min}\exp\left(-\frac{4\lambda_{\min}}{\kappa_{\max}^2(C_{\max}+1)^2}\right)+(1-\lambda_{\min})$ and the last inequality comes from the fact that the function $f(x)\triangleq x\exp\left(-\frac{4x}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)+(1-x)$ is non-increasing in $x\in[0,1]$. Thus,

$$
\begin{aligned}
\sum_{t=K+2}^{T}\mathbb{E}\left[A_t^{j,2}\right] &\leq 2\sum_{t=K+2}^{T}\left(\exp\left(-\frac{2\lambda_{\min}^2(t-K-1)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\rho_2^K\right)\\
&\leq 2\cdot\frac{\exp\left(\frac{-2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)}{1-\exp\left(\frac{-2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_2^K\\
&= \frac{2}{\exp\left(\frac{2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)-1}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_2^K\\
&\leq \frac{2}{1+\frac{2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}-1}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_2^K\\
&= \frac{\kappa_{\max}^2\cdot(C_{\max}+1)^2}{\lambda_{\min}^2}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_2^K, \qquad \text{(EC.22)}
\end{aligned}
$$

where the last inequality comes from the fact that $e^x\geq 1+x$ for $x\geq 0$.

Therefore, by (EC.20) and (EC.22), (EC.15) is upper bounded by

$$
\begin{aligned}
&a_{\max}\cdot M\cdot r_{\max}\cdot\sum_{t=K+2}^{T}\sum_{j\in[N]}\lambda_j\cdot\mathbb{P}\left(X_{t,s_t^{\text{on}}}^*<a_{\max}+1|\Theta_t=j\right)\\
&\leq a_{\max}\cdot M\cdot r_{\max}\cdot\sum_{j\in[N]}\lambda_j\left(\left\lceil\frac{C_1}{\lambda_{\min}}\right\rceil\cdot\rho_1^{K-2\frac{C_1}{\lambda_{\min}}}+\frac{\kappa_{\max}^2\cdot(C_{\max}+1)^2}{\lambda_{\min}^2}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_2^K\right)\\
&\leq a_{\max}\cdot M\cdot r_{\max}\cdot\left(\left\lceil\frac{C_1}{\lambda_{\min}}\right\rceil\cdot\rho_1^{K-2\frac{C_1}{\lambda_{\min}}}+\frac{\kappa_{\max}^2\cdot(C_{\max}+1)^2}{\lambda_{\min}^2}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_2^K\right)\\
&\leq a_{\max}\cdot M\cdot r_{\max}\cdot\left(\left\lceil\frac{C_1}{\lambda_{\min}}\right\rceil\cdot\rho_1^{-2\frac{C_1}{\lambda_{\min}}}+\frac{\kappa_{\max}^2\cdot(C_{\max}+1)^2}{\lambda_{\min}^2}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\right)\cdot\rho^K,
\end{aligned}
$$

where $\rho=\max\{\rho_1,\rho_2\}$. The proof of Theorem 2 is completed. □

EC.4. Proofs of Theorems 3 and 4 in Section 6

Proof of Theorem 3. Recall that the estimated arrival probability of type $j \in [N]$ in period t is

$$\hat{\lambda}_{t,j} = \frac{D_j[T, t-K]}{T-t+1+K}.$$

Additionally define the following event:

$$F_{t,j} \triangleq \left\{ \left| \hat{\lambda}_{t,j} - \lambda_j \right| \le \eta \right\},$$

where $\eta = \frac{\lambda_{\min}}{3} \min\left\{ \frac{1}{\kappa_{\max}(C_{\max}+1)},\ 1 \right\}$ as defined in (26). Let $F_{t,j}^c$ be the complement set of $F_{t,j}$. By Hoeffding's inequality, for all $j \in [N]$,

$$\mathbb{P}\left(F_{t,j}^c\right) = \mathbb{P}\Big(\left| D_j[T, t-K] - \lambda_j(T-t+1+K) \right| > \eta(T-t+1+K) \Big) \le 2e^{-2\eta^2(T-t+1+K)}. \tag{EC.23}$$

Similar to (21) and (EC.15), we have

$$X_{t,s_t^{\text{on}}} \ge \frac{D_{\Theta_t}[t, t-K] + \hat{\lambda}_{t,\Theta_t}[t-K-1, 1]}{C_{\max}+1},$$

and

$$\text{REG} \le a_{\max} \cdot M \cdot r_{\max} \cdot \sum_{t=K+2}^{T} \sum_{j \in [N]} \lambda_j \cdot \mathbb{P}\left(X_{t,s_t^{\text{on}}}^* < a_{\max}+1 | \Theta_t = j \right). \tag{EC.24}$$

We next bound $\mathbb{P}\left(X_{t,s_t^{\text{on}}}^* < a_{\max}+1 | \Theta_t = j \right)$. Similar to (EC.16),

$$\begin{aligned}
&\mathbb{P}\left(X_{t,s_t^{\text{on}}}^* < a_{\max}+1 \middle| \Theta_t = j \right) \\
&\le \mathbb{P}\left(\left| D_{j_t^*}[t-K-1, 1] - \hat{\lambda}_{t,j_t^*}(t-K-1) \right| > \frac{D_j[t-1, t-K] + \hat{\lambda}_{t,j}(t-K-1) - C_1}{\kappa_{\max}(C_{\max}+1)} \middle| \Theta_t = j \right) \\
&\triangleq \mathbb{P}(A) \\
&= \mathbb{P}\left(A,\ F_{t,j_t^*}^c\right) + \mathbb{P}\left(A,\ F_{t,j_t^*},\ F_{t,j}^c\right) + \mathbb{P}\left(A,\ F_{t,j_t^*},\ F_{t,j}\right) \\
&= \mathbb{P}\left(A,\ F_{t,j_t^*}^c\right) + \mathbb{P}\left(A,\ F_{t,j_t^*},\ F_{t,j}^c\right) + \mathbb{P}\left(F_{t,j_t^*},\ F_{t,j}\right) \cdot \mathbb{P}\left(A \middle| F_{t,j_t^*},\ F_{t,j}\right) \\
&\le \mathbb{P}\left(F_{t,j_t^*}^c\right) + \mathbb{P}\left(F_{t,j}^c\right) + \mathbb{P}\left(A \middle| F_{t,j_t^*},\ F_{t,j}\right),
\end{aligned}$$

where A represents the conditional event

$$\left\{ \left| D_{j_t^*}[t-K-1, 1] - \hat{\lambda}_{t,j_t^*}(t-K-1) \right| > \frac{D_j[t-1, t-K] + \hat{\lambda}_{t,j}(t-K-1) - C_1}{\kappa_{\max}(C_{\max}+1)} \middle| \Theta_t = j \right\}.$$

From (EC.23),

$$\mathbb{P}\left(F^c_{t,j^*_t}\right) \leq \ 2e^{-2\eta^2(T-t+1+K)}, \qquad \mathbb{P}\left(F^c_{t,j}\right) \leq \ 2e^{-2\eta^2(T-t+1+K)}.$$

In addition, it is straightforward to verify that conditional on F_{t,j^*_t},

$$\begin{aligned} &\left|D_{j^*_t}[t-K-1,1] - \hat{\lambda}_{t,j^*_t}(t-K-1)\right| \\ \leq &\left|D_{j^*_t}[t-K-1,1] - \lambda_{j^*_t}(t-K-1)\right| + \left|\left(\hat{\lambda}_{t,j^*_t} - \lambda_{j^*_t}\right)(t-K-1)\right| \\ \leq &\left|D_{j^*_t}[t-K-1,1] - \lambda_{j^*_t}(t-K-1)\right| + \eta(t-K-1), \end{aligned} \tag{EC.25}$$

and conditional on $F_{t,j}$,

$$\hat{\lambda}_{t,j} \geq \ \lambda_j - \eta \geq \ \eta \cdot \kappa_{\max}(C_{\max}+1) + \frac{\lambda_j}{3}. \tag{EC.26}$$

(EC.25) and (EC.26) together imply that

$$\begin{aligned} &\mathbb{P}\left(A \middle| F_{t,j^*_t},\ F_{t,j}\right) \\ \leq &\ \mathbb{P}\left(\left|D_{j^*_t}[t-K-1,1] - \lambda_{j^*_t}(t-K-1)\right| > \frac{D_j[t-1,t-K] + \lambda_j(t-K-1)/3 - C_1}{\kappa_{\max}(C_{\max}+1)}\right). \end{aligned}$$

Moreover, similar to (EC.20) and (EC.22), we can derive

$$\begin{aligned} &\sum_{t=K+2}^{T} \mathbb{P}\left(\left|D_{j^*_t}[t-K-1,1] - \lambda_{j^*_t}(t-K-1)\right| > \frac{D_j[t-1,t-K] + \lambda_j(t-K-1)/3 - C_1}{\kappa_{\max}(C_{\max}+1)}\right) \\ \leq &\ \left\lceil \frac{3C_1}{\lambda_{\min}} \right\rceil \cdot \hat{\rho}_1^{K-2\frac{C_1}{\lambda_{\min}/3}} + \frac{\kappa^2_{\max}\cdot(C_{\max}+1)^2}{(\lambda_{\min}/3)^2} \cdot \exp\left(\frac{4\cdot(\lambda_{\max}/3)\cdot C_1}{\kappa^2_{\max}\cdot(C_{\max}+1)^2}\right) \cdot \hat{\rho}_2^K, \end{aligned}$$

where $\hat{\rho}_1$ and $\hat{\rho}_2$ are defined in (26). Combining all the above, (EC.24) is upper bounded by

$$\begin{aligned} &a_{\max}\cdot M\cdot r_{\max}\cdot \sum_{t=K+2}^{T}\sum_{j\in[N]} \lambda_j \cdot \mathbb{P}\left(X^*_{t,s^{\text{on}}_t} < a_{\max}+1 | \Theta_t = j\right) \\ \leq &\ a_{\max}\cdot M\cdot r_{\max}\cdot \sum_{t=K+2}^{T} 4e^{-2\eta^2(T-t+1+K)} \\ &+ a_{\max}\cdot M\cdot r_{\max}\cdot \left(\left\lceil \frac{3C_1}{\lambda_{\min}} \right\rceil \cdot \hat{\rho}_1^{K-2\frac{C_1}{\lambda_{\min}/3}} + \frac{\kappa^2_{\max}\cdot(C_{\max}+1)^2}{(\lambda_{\min}/3)^2} \cdot \exp\left(\frac{4\cdot(\lambda_{\max}/3)\cdot C_1}{\kappa^2_{\max}\cdot(C_{\max}+1)^2}\right) \cdot \hat{\rho}_2^K\right) \\ \leq &\ a_{\max}\cdot M\cdot r_{\max}\cdot \left(\frac{2}{\eta^2} + \left\lceil \frac{3C_1}{\lambda_{\min}} \right\rceil \cdot \hat{\rho}_1^{-2\frac{C_1}{\lambda_{\min}/3}} + \frac{\kappa^2_{\max}\cdot(C_{\max}+1)^2}{(\lambda_{\min}/3)^2} \cdot \exp\left(\frac{4\cdot(\lambda_{\max}/3)\cdot C_1}{\kappa^2_{\max}\cdot(C_{\max}+1)^2}\right)\right) \cdot \hat{\rho}^K, \end{aligned}$$

where the last inequality comes from the fact that

$$\sum_{t=K+2}^{T} 4e^{-2\eta^2(T-t+1+K)} \leq \ 4e^{-2\eta^2 K}\frac{e^{-2\eta^2}}{1-e^{-2\eta^2}} \leq \ \frac{2}{\eta^2}\cdot \hat{\rho}_3^K.$$

The proof of Theorem 3 is completed. □

Proof of Theorem 4. As discussed in Section 6.3, $T = (K+1)(\lceil T/(K+1)\rceil - 1) + R$ and there are $\lceil T/(K+1)\rceil$ batches. It suffices to consider the regret incurred in the first $\lceil T/(K+1)\rceil - 1$ batches. Particularly, the n-th batch consists of $\Theta_{T-(n-1)(K+1)}, \Theta_{T-(n-1)(K+1)-1}, \ldots, \Theta_{T-n(K+1)+1}$ for $n = 1, \ldots, \lceil T/(K+1)\rceil - 1$, and the decision for $\Theta_{T-(n-1)(K+1)-(K-m)}$ in the n-th batch is equivalent to m-period delay based on the expected demand $\lambda[T - n(K+1), 1]$ for $m = K, K-1, \ldots, 0$. Each arrival in the first $\lceil T/(K+1)\rceil - 1$ batches can be specified by some $n \in [\lceil T/(K+1)\rceil - 1]$ and $m \in \{0\} \cup [K]$:

$$\begin{aligned} t = \ t(n,m) &\triangleq T - (n-1)(K+1) - (K-m) \\ &= (K+1)\left(\lceil T/(K+1)\rceil - 1 - n\right) + (m+1) + R. \end{aligned} \tag{EC.27}$$

Namely, the arrival at time $t(n,m)$ is in the n-th batch with m-period delay.

By the characterization of (n,m) and (EC.15), we have

$$\text{REG} \leq \ a_{\max} \cdot M \cdot r_{\max} \cdot \sum_{n=1}^{\lceil T/(K+1)\rceil - 1} \sum_{m=0}^{K} \sum_{j \in [N]} \lambda_j \cdot \mathbb{P}\left(X^*_{t(n,m), s^{\text{on}}_{t(n,m)}} < a_{\max} + 1 | \Theta_{t(n,m)} = j \right).$$

Similar to (EC.16), $\mathbb{P}\left(X^*_{t(n,m), s^{\text{on}}_{t(n,m)}} < a_{\max} + 1 | \Theta_{t(n,m)} = j \right)$ can be bounded by

$$\mathbb{P}\left(X^*_{t(n,m), s^{\text{on}}_{t(n,m)}} < a_{\max} + 1 | \Theta_{t(n,m)} = j \right) \leq \ \mathbb{P}\left(E^j_{t(n,m)} \right),$$

where $E^j_{t(n,m)}$ represents the following event conditional on $\Theta_{t(n,m)} = j$:

$$\begin{aligned} &\left| D_{j^*_{t(n,m)}}[T - n(K+1), 1] - \lambda_{j^*_{t(n,m)}}(T - n(K+1)) \right| \\ > \ &\frac{D_j[t(n,m) - 1, t(n,m) - m] + \lambda_j(T - n(K+1)) - C_1}{\kappa_{\max}(C_{\max} + 1)}. \end{aligned}$$

Here, $j^*_{t(n,m)} \in \arg\max_{j \in [N]} \{ |D_j[T - n(K+1), 1] - \lambda_j(T - n(K+1))| \}$ and $t(n,m) - m - 1 = T - n(K+1)$ from (EC.27). Similar to (EC.17), define

$$\mathbb{P}\left(E^j_{t(n,m)} | \Theta_{t(n,m)-1}, \ldots, \Theta_{t(n,m)-m} \right) = \ A^{j,1}_{t(n,m)} + A^{j,2}_{t(n,m)},$$

where

$$\begin{aligned} A^{j,1}_{t(n,m)} &\triangleq \mathbb{I}\left(D_j[t(n,m) - 1, t(n,m) - m] + \lambda_j(T - n(K+1)) - C_1 < 0 \right) \\ &\quad \cdot \mathbb{P}\left(E^j_{t(n,m)} \middle| D_j[t(n,m) - 1, t(n,m) - m] + \lambda_j(T - n(K+1)) - C_1 < 0 \right), \\ A^{j,2}_{t(n,m)} &\triangleq \mathbb{I}\left(D_j[t(n,m) - 1, t(n,m) - m] + \lambda_j(T - n(K+1)) - C_1 \geq 0 \right) \\ &\quad \cdot \mathbb{P}\left(E^j_{t(n,m)} \middle| D_j[t(n,m) - 1, t(n,m) - m] + \lambda_j(T - n(K+1)) - C_1 \geq 0 \right). \end{aligned}$$

It is easy to see $\mathbb{P}\left(E^j_{t(n,m)}\middle| D_j[t(n,m)-1,t(n,m)-m]+\lambda_j(T-n(K+1))-C_1<0\right)=1$. It implies that

$$\begin{aligned}
&\sum_{n=1}^{\lceil T/(K+1)\rceil-1}\sum_{m=0}^{K}\mathbb{E}\left[A^{j,1}_{t(n,m)}\right]\\
=&\sum_{n=1}^{\lceil T/(K+1)\rceil-1}\sum_{m=0}^{K}\mathbb{P}\left(D_j[t(n,m)-1,t(n,m)-m]+\lambda_j(T-n(K+1))-C_1<0\right)\cdot 1\\
=&\sum_{m=0}^{K}\sum_{n=1}^{\lceil T/(K+1)\rceil-1}\mathbb{P}\left(\mathrm{Bin}(m,\lambda_j)<C_1-\lambda_j(T-n(K+1))\right)\\
=&\sum_{m=0}^{K}\sum_{n=0}^{\lceil T/(K+1)\rceil-2}\mathbb{P}\left(\mathrm{Bin}(m,\lambda_j)<C_1-\lambda_j(n(K+1)+R)\right)\\
\le&\sum_{m=0}^{K}\min\left\{\left\lceil\frac{T}{K+1}\right\rceil-1,\ \left\lceil\frac{C_1}{\lambda_{\min}(K+1)}\right\rceil\right\}\mathbb{P}\left(\mathrm{Bin}(m,\lambda_j)<C_1-\lambda_jR\right),
\end{aligned}$$

where the last equality comes from the expression of $T=(K+1)(\lceil T/(K+1)\rceil-1)+R$ and the last inequality is similar to (EC.18). Thus, by applying Hoeffding's inequality and using a similar argument to (EC.19),

$$\sum_{n=1}^{\lceil T/(K+1)\rceil-1}\sum_{m=0}^{K}\mathbb{E}\left[A^{j,1}_{t(n,m)}\right]\le\min\left\{\left\lceil\frac{T}{K+1}\right\rceil-1,\ \left\lceil\frac{C_1}{\lambda_{\min}(K+1)}\right\rceil\right\}\sum_{m=0}^{K}\rho_1^{m+2R-2\frac{C_1}{\lambda_{\min}}},\tag{EC.28}$$

where $\rho_1=\exp(-2\lambda_{\min}^2)$.

Next, we bound $\sum_{n=1}^{\lceil T/(K+1)\rceil-1}\sum_{m=0}^{K}\mathbb{E}\left[A^{j,2}_{t(n,m)}\right]$ from above: by Hoeffding's inequality,

$$\begin{aligned}
&\mathbb{P}\left(E^j_{t(n,m)}\middle| D_j[t(n,m)-1,t(n,m)-m]+\lambda_j(T-n(K+1))-C_1\ge 0\right)\\
\le\ &2\exp\left(-2\frac{\left[D_j[t(n,m)-1,t(n,m)-m]+\lambda_j(T-n(K+1))-C_1\right]^2}{(T-n(K+1))\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right),
\end{aligned}$$

which implies

$$\mathbb{E}\left[A^{j,2}_{t(n,m)}\right]\le\ 2\mathbb{E}\left[\exp\left(-2\frac{\left[D_j[t(n,m)-1,t(n,m)-m]+\lambda_j(T-n(K+1))-C_1\right]^2}{(T-n(K+1))\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\right].$$

In addition, similar to (EC.21),

$$\begin{aligned}
&\mathbb{E}\left[\exp\left(-2\frac{\left[D_j[t(n,m)-1,t(n,m)-m]+\lambda_j(T-n(K+1))-C_1\right]^2}{(T-n(K+1))\cdot\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\right]\\
\le\ &\exp\left(-\frac{2\lambda_{\min}^2(T-n(K+1))}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\rho_2^m,
\end{aligned}$$

where $\rho_2 = \lambda_{\min}\exp\left(-\frac{4\lambda_{\min}}{\kappa_{\max}^2(C_{\max}+1)^2}\right) + (1-\lambda_{\min})$. Thus, we have

$$
\begin{aligned}
&\sum_{n=1}^{\lceil T/(K+1)\rceil-1} \mathbb{E}\left[A_{t(n,m)}^{j,2}\right] \\
\le\ & 2\sum_{n=1}^{\lceil T/(K+1)\rceil-1}\left(\exp\left(-\frac{2\lambda_{\min}^2(T-n(K+1))}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\rho_2^m\right) \\
=\ & 2\left(\sum_{n=0}^{\lceil T/(K+1)\rceil-2}\exp\left(-\frac{2\lambda_{\min}^2(n(K+1)+R)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\right)\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\rho_2^m \\
=\ & \frac{2\left(1-\exp\left(-\frac{2\lambda_{\min}^2(K+1)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\left(\lceil\frac{T}{K+1}\rceil-1\right)\right)\right)\exp\left(-\frac{2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}R\right)}{1-\exp\left(-\frac{2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}(K+1)\right)}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_2^m \\
\le\ & \frac{2\left(1-\exp\left(-\frac{2\lambda_{\min}^2(K+1)}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\left(\lceil\frac{T}{K+1}\rceil-1\right)\right)\right)}{1-\exp\left(-\frac{2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_3^R\cdot\rho_2^m \\
=\ & \frac{2\left(1-\rho_3^{(K+1)\left(\lceil\frac{T}{K+1}\rceil-1\right)}\right)}{1-\rho_3}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\rho_3^R\cdot\rho_2^m, \qquad \text{(EC.29)}
\end{aligned}
$$

where $\rho_3 = \exp\left(-\frac{2\lambda_{\min}^2}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)$.

Therefore, by combining (EC.28) and (EC.29), we conclude that

$$
\begin{aligned}
&\sum_{n=1}^{\lceil T/(K+1)\rceil-1}\sum_{m=0}^{K}\sum_{j\in[N]}\lambda_j\cdot\mathbb{P}\left(X^*_{t(n,m),s_{t(n,m)}} < a_{\max}+1 \,|\, \Theta_{t(n,m)}=j\right) \\
\le\ & \min\left\{\left\lceil\frac{T}{K+1}\right\rceil-1,\ \left\lceil\frac{C_1}{\lambda_{\min}(K+1)}\right\rceil\right\}\cdot\sum_{m=0}^{K}\rho_1^{m-2\frac{C_1}{\lambda_{\min}}}\cdot\rho_1^{2R} \\
&+\frac{2\left(1-\rho_3^{(K+1)\left(\lceil\frac{T}{K+1}\rceil-1\right)}\right)}{1-\rho_3}\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)\cdot\sum_{m=0}^{K}\rho_2^m\cdot\rho_3^R \\
\le\ & \left(c_1\cdot\sum_{m=0}^{K}\rho_1^{m-2\frac{C_1}{\lambda_{\min}}}+\frac{2c_2\cdot\exp\left(\frac{4\lambda_{\max}\cdot C_1}{\kappa_{\max}^2\cdot(C_{\max}+1)^2}\right)}{1-\rho_3}\cdot\sum_{m=0}^{K}\rho_2^m\right)\cdot\rho^R,
\end{aligned}
$$

where $c_1 = \min\left\{\lceil\frac{T}{K+1}\rceil-1,\ \left\lceil\frac{C_1}{\lambda_{\min}(K+1)}\right\rceil\right\}$, $c_2 = 1-\rho_3^{(K+1)\left(\lceil\frac{T}{K+1}\rceil-1\right)}$ and $\rho = \max\{\rho_1^2, \rho_3\}$. The proof of Theorem 4 is completed. □